

THE PROBLEM

Switch on a **light** tonight

94%

Natural gas in Singapore’s electricity fuel mix, 2024.

Source: EMA, Our Energy Story; EMA, Singapore Energy Statistics 2025



THE PROBLEM

Hydrogen could carry **up to half**

Up to

50%

Low-carbon hydrogen, a possible share of power needs by 2050.

Source: METI, Hydrogen



THE PROBLEM

Today’s hydrogen carries a **carbon bill**

9.21

kg CO_2 per kg H_2

Direct emissions of an unabated steam methane reformer, INL basis.

Source: INL TEV-953 Table 2, PDF p30. Status: reduced-order balances anchored on INL cases.



PROJECT

NUS-H2

Nuclear-Upgraded Steam Methane Reforming with Carbon Capture for Low-Carbon Hydrogen in Singapore.

Wee Zhern Ray

SCOPE AND LIMITS

Two targets, one straight answer

Net abatement

met

Result 0.578 Mt CO_2e/y, target above 0.250

Abatement cost

missed

Result S$165.6 to S$277.9/t, target under S$100

0.250

Mt CO_2e/y net abatement

S$100/t

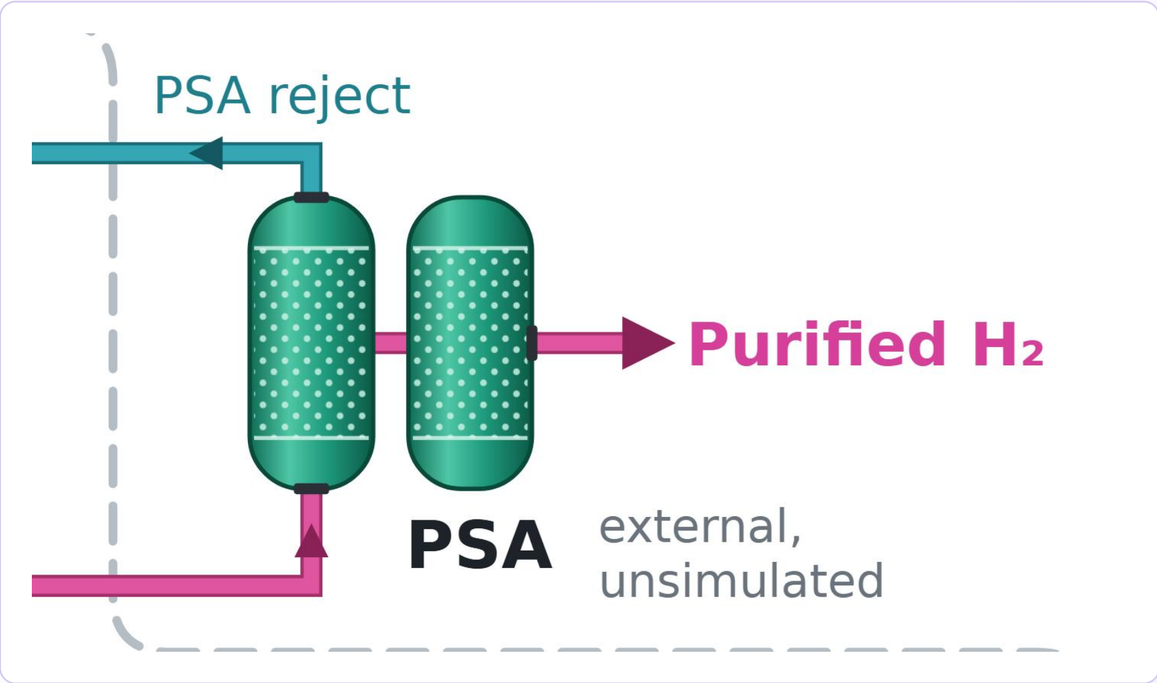
abatement cost

Abatement target: met. Cost target: missed.

Source: Project statement. Status: reduced-order balances anchored on INL cases.

SCOPE AND LIMITS

One plant at **real industrial scale**



100,000

Nm^3/h of purified hydrogen

66,925 t H_2/y at 85% availability. PSA reject returns across the boundary.

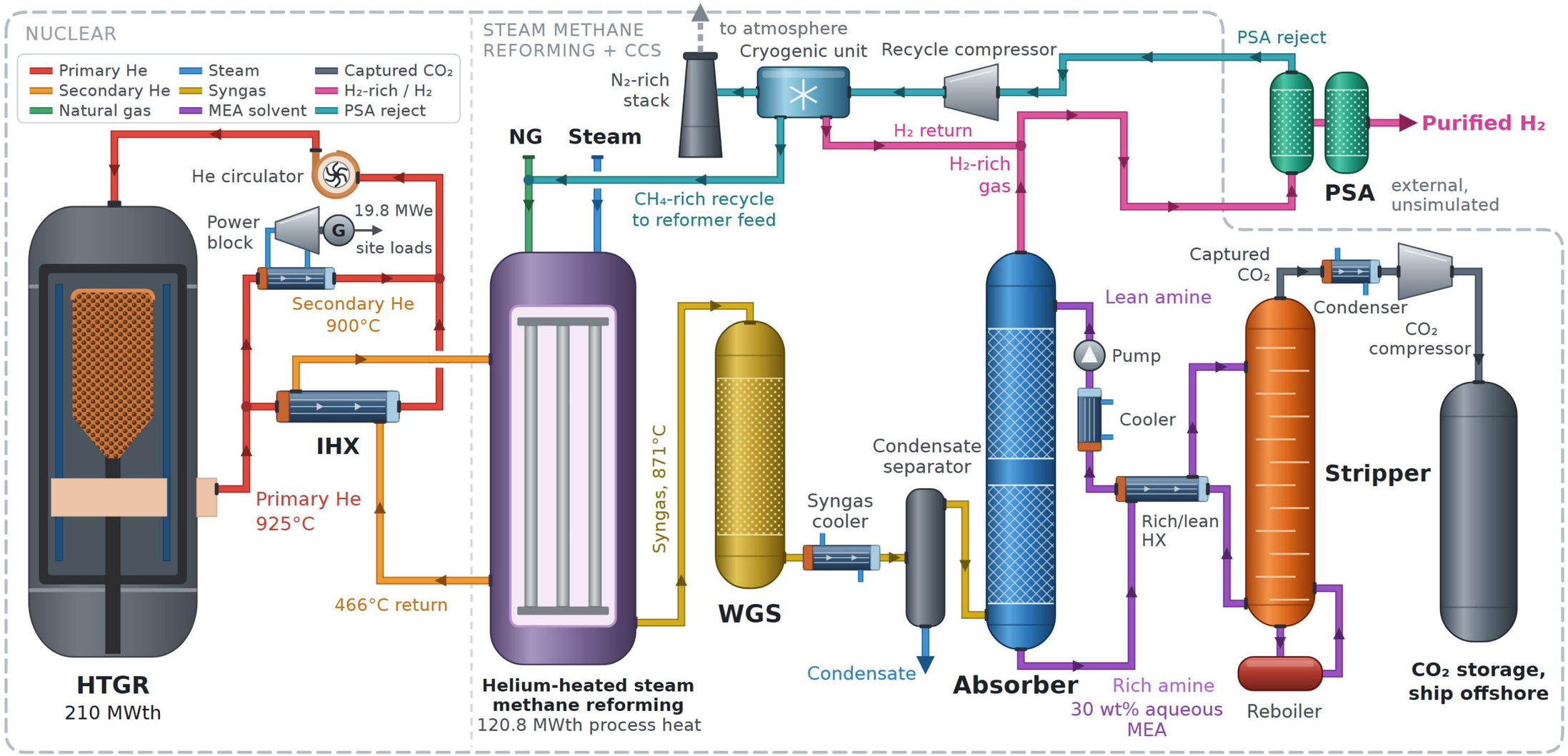
Source: Air Liquide 2008; Air Liquide 2011; Decisions D04, D05. Status: reduced-order balances anchored on INL cases.

PROPOSED SOLUTION

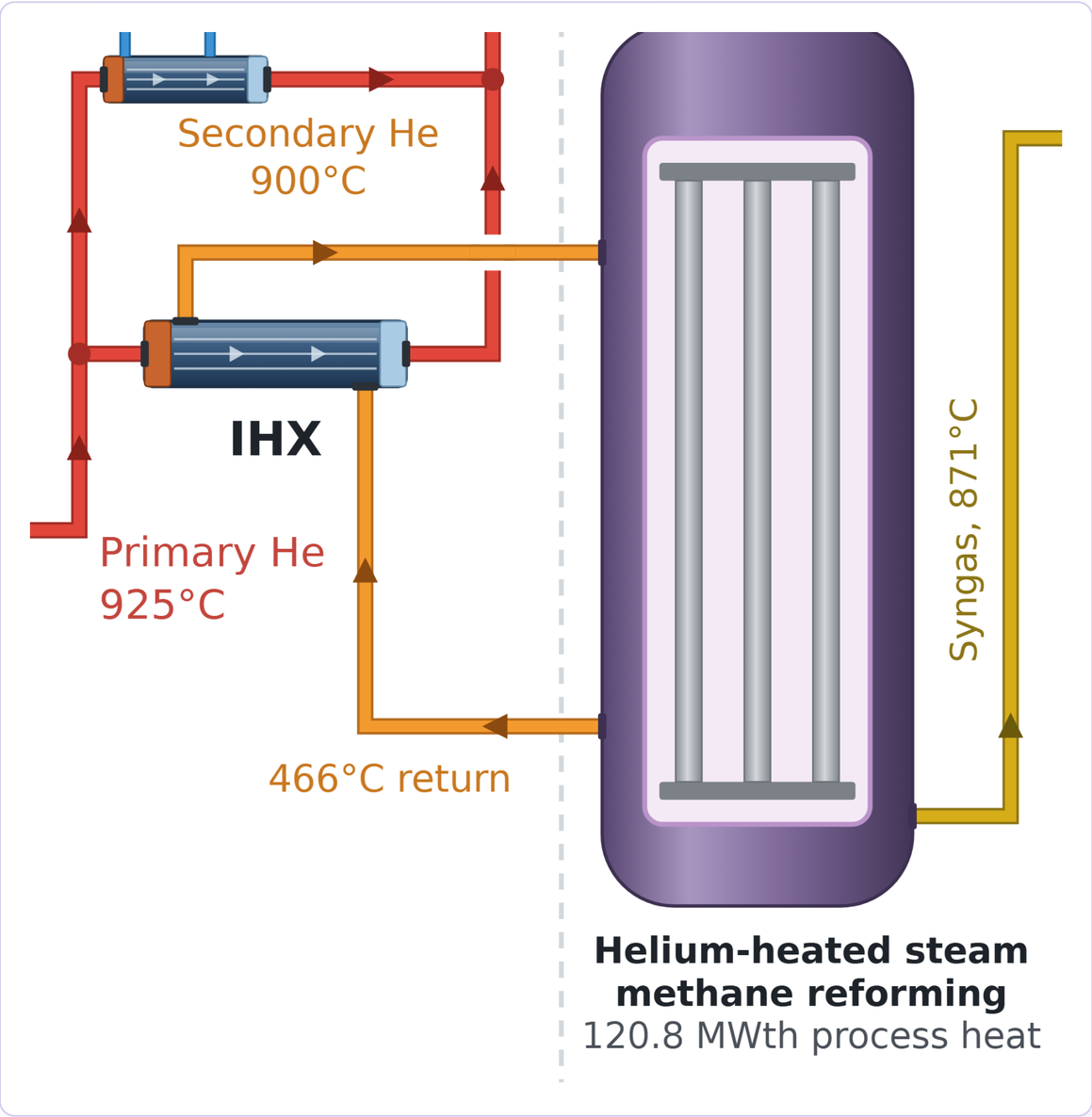
A reformer heated by **helium**

Generic HTGR, helium-heated steam methane reforming, MEA capture, ship offshore storage.

210 MWth
Reactor rating



Source: INL TEV-961 Table 2; Decisions D06, D23. Status: reduced-order balances anchored on INL cases.



PROPOSED SOLUTION

Helium heat **replaces the furnace**

120.8

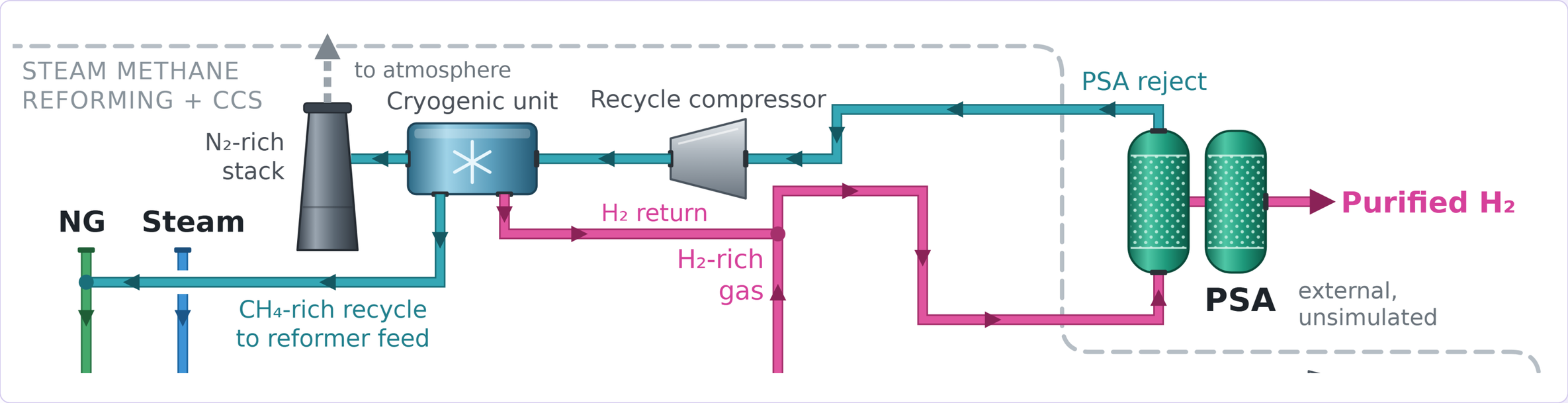
MWth at 900 °C

Covers reforming and preheat. Helium returns at 466 °C.

Source: INL TEV-961 printed p12 to p13, p21, PDF p14. Status: reduced-order balances anchored on INL cases.

PROPOSED SOLUTION

Less gas for the same hydrogen



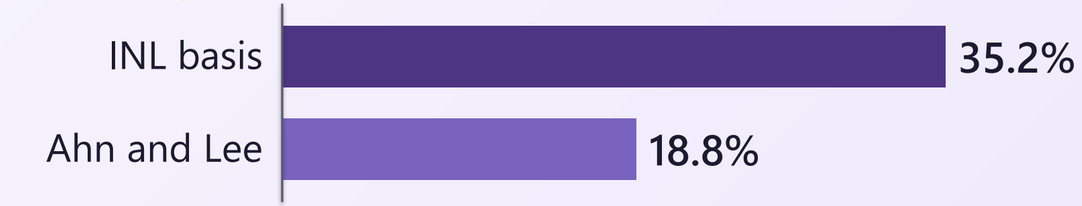
23.2

MMSCFD, NUS-H2

35.9

MMSCFD, unabated steam methane reforming

Gas saving from published flows

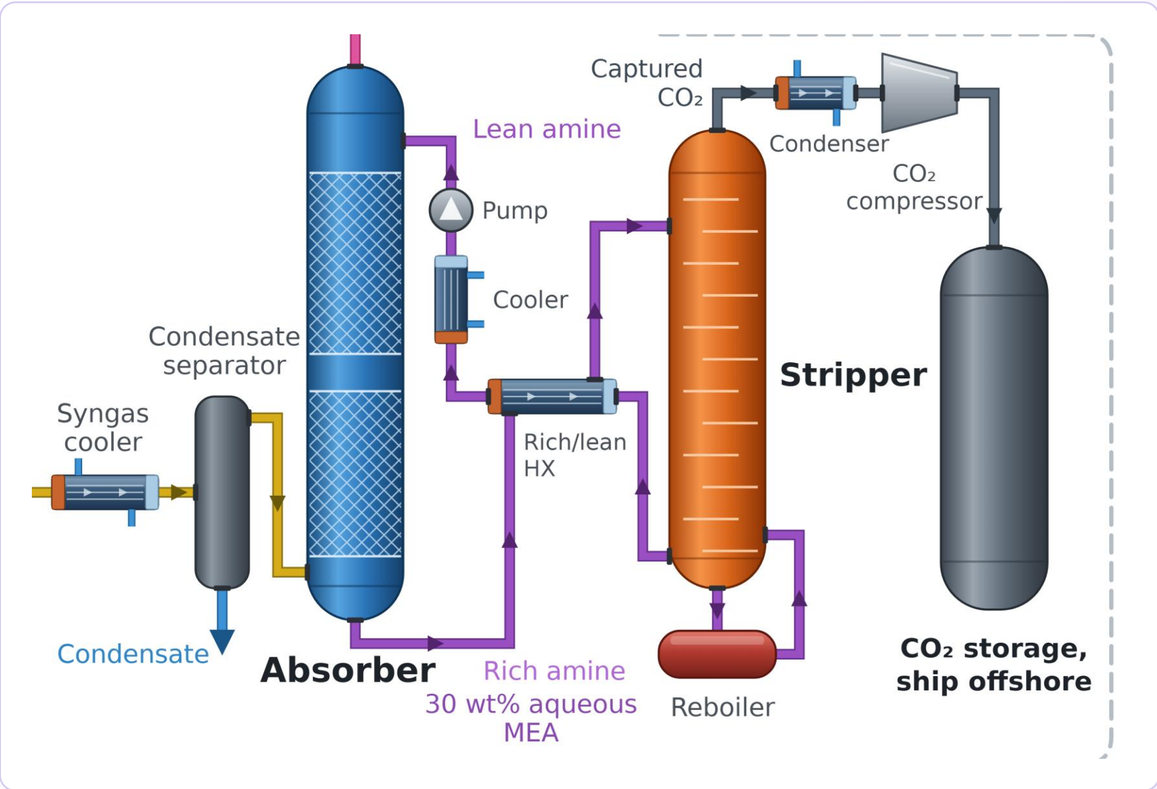


12.6 MMSCFD saved. Depends on recycle transferring from Selexol to MEA.

Source: INL TEV-953 Table 2, PDF p30; Ahn and Lee 2026; Decision D33. Status: reduced-order balances anchored on INL cases.

PROPOSED SOLUTION

Capture, **counted two ways**



90%

per pass

90.4%

of plant carbon

30 wt% MEA. 0.360 Mt CO_2/y to storage. Per-pass value is a project assumption.

Source: Decisions D07, D25; IEAGHG 2022-07 PDF p44. Status: reduced-order balances anchored on INL cases.

PROPOSED SOLUTION

In INL's own words

> Increasing the HTGR ROT significantly reduces natural gas consumption

R. Wood, INL TEV-961, 2010, Executive Summary

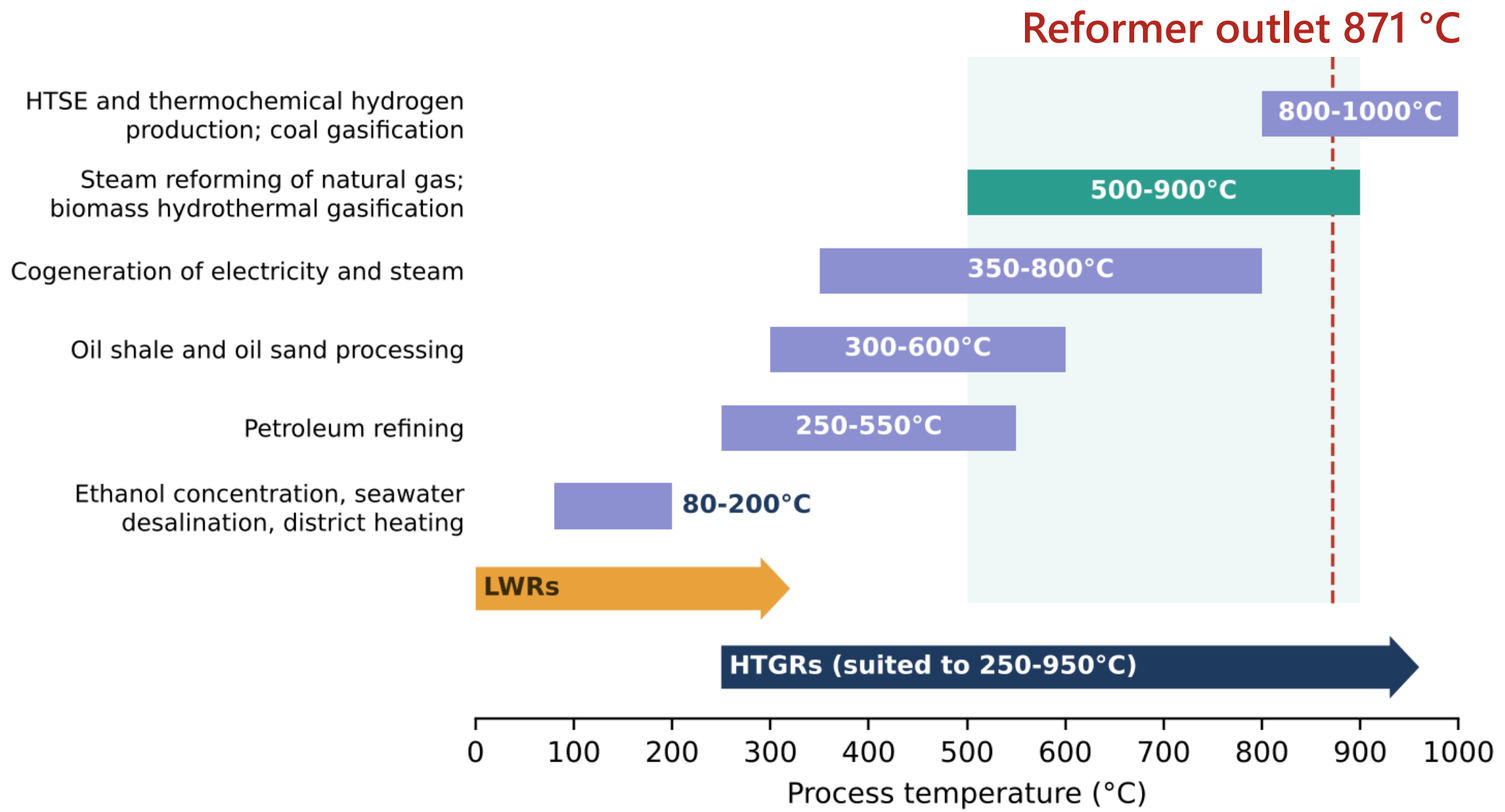
30.9%

less gas than INL's conventional plant with capture, at 925 °C

INL has not reviewed NUS-H2.

Source: INL TEV-961 Executive Summary, PDF p5

Why nuclear: the **temperature fits**



925 °C

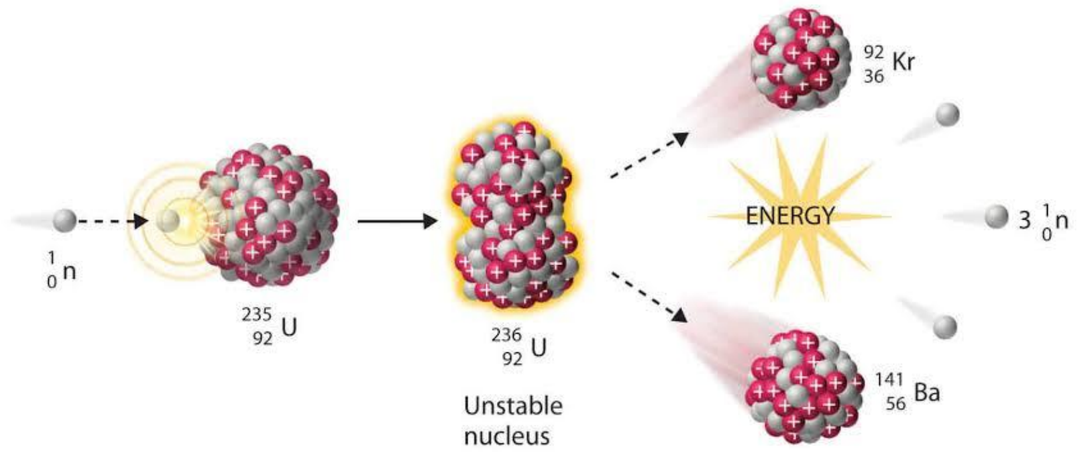
reactor outlet

Reformer outlet 871 °C. Only test reactors have run this hot.

Source: Mears 2010; Nelson et al. 2011 PDF p9, p21; IAEA NP-T-4.3 printed p23; JAEA, HTTR; WNN 2023

WHY NUCLEAR

Heat and power from one reactor



Site electrical load, MWe

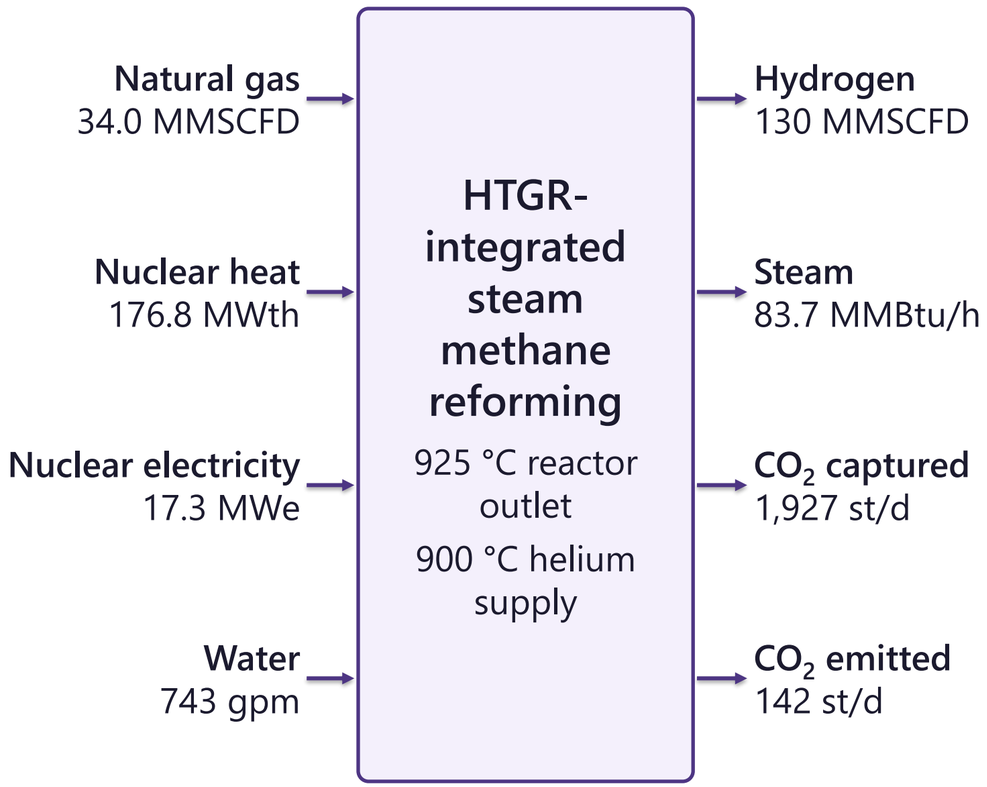
12.33	7.45
Process loads	Helium circulators

19.8

MWe site load, grid import 0

Process loads 12.33 MWe. Helium circulators add 7.45 MWe.

Source: Nuclear-Power.com; Decision D24; Integrated_Energy_Balance.csv. Status: reduced-order balances anchored on INL cases.



Case A, INL reference (Wood, 2010)

BASIS AND SIZING

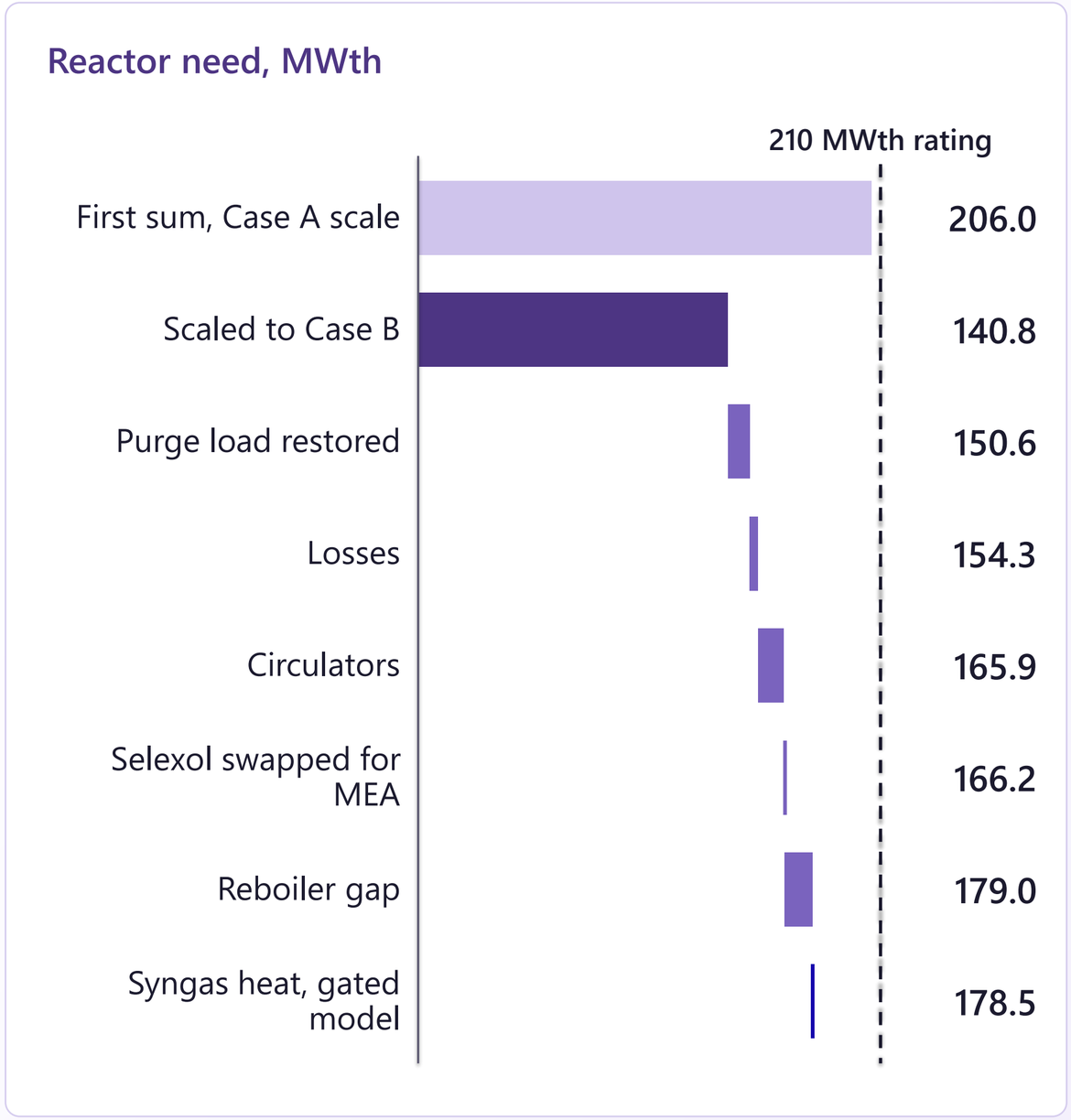
INL reference: Case A

130

MMSCFD hydrogen, INL reference

INL reference. Separate from the proposed 210 MWth design.

Source: INL TEV-961 Table 2; Decision D03. Status: reduced-order balances anchored on INL cases.



BASIS AND SIZING

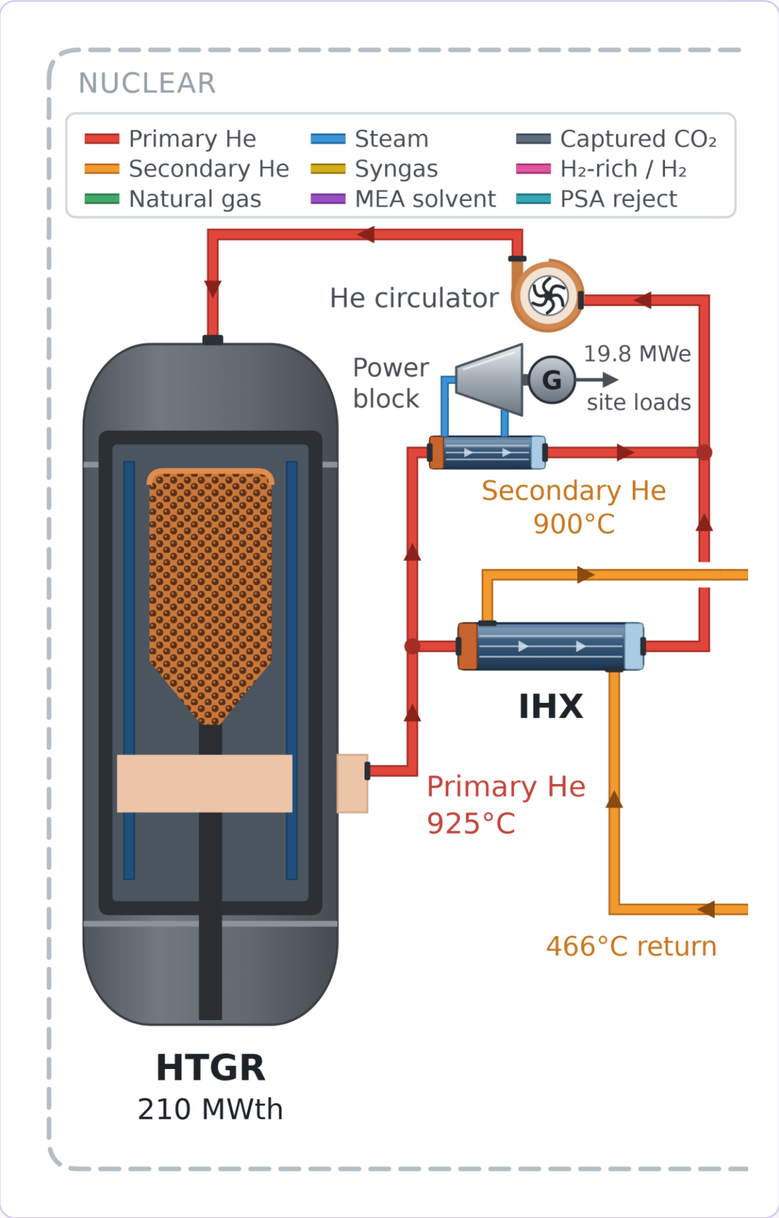
The walk to 178.5 MWth

178.5

MWth needed, range 155.8 to 214.1

SRD of 3.8 GJ/t is an assumption with no syngas measurement.

Source: Design decisions section 5; Decisions D24, D26; Validation_Results.csv sizing walk rows.
Status: reduced-order balances anchored on INL cases.



BASIS AND SIZING

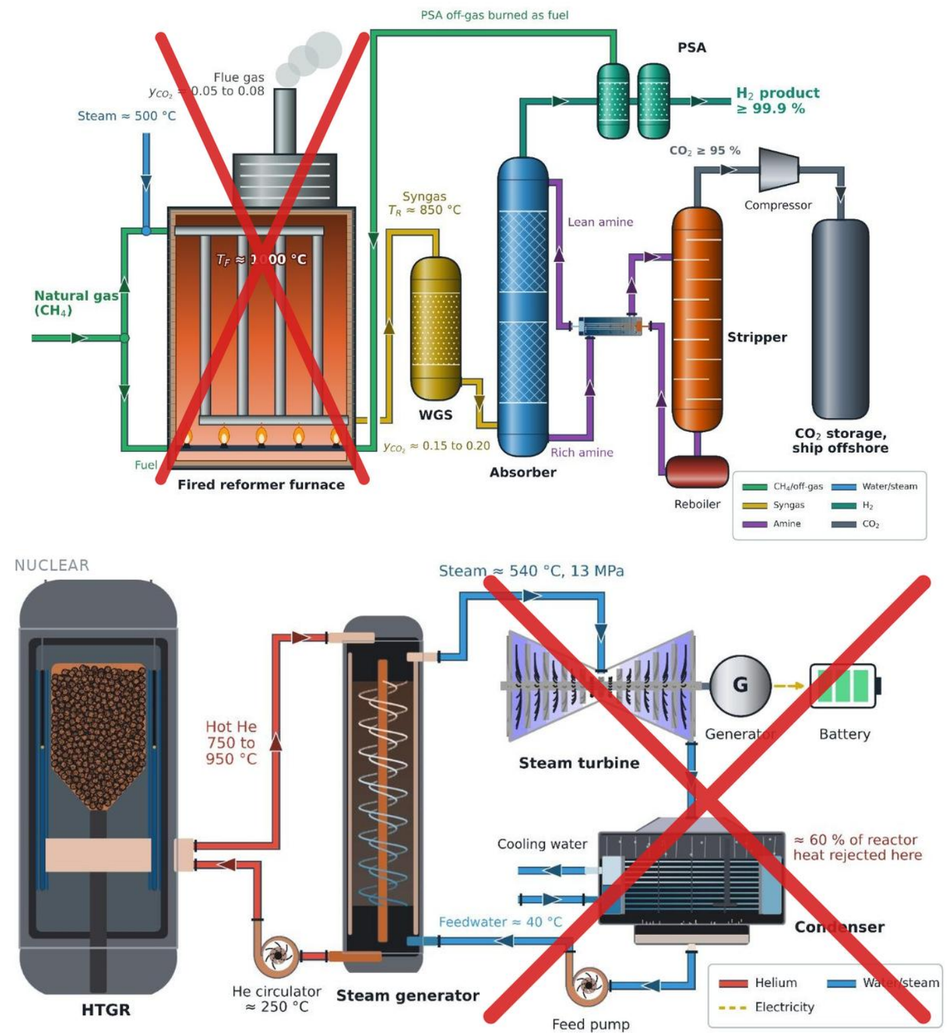
A margin that **earns nothing**

31.5

MWth spare, no revenue

6.5 MWth on Ahn and Lee’s heat basis. 4.1 MWth short at the adverse corner.

Source: Decisions D03, X6, X7; Ahn and Lee 2026 Table 6. Status: reduced-order balances anchored on INL cases.



Condensing power plant removed. A small power block stays for site loads.

BENEFITS OF INTEGRATION

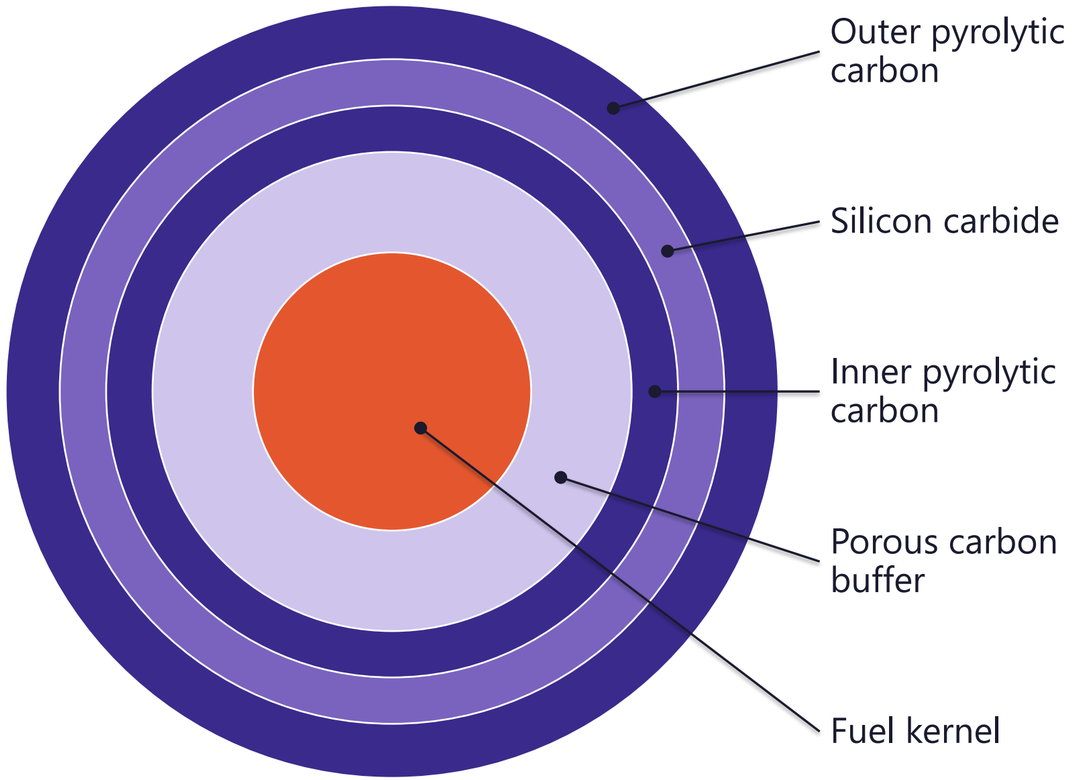
Integration **removes equipment**

218,274

t CO_2/y never formed

The fired furnace goes. So does the large condensing power plant.

Source: INL TEV-961; Decision X12; Emissions_Ledger.csv. Status: reduced-order balances anchored on INL cases.



RISKS: SAFETY

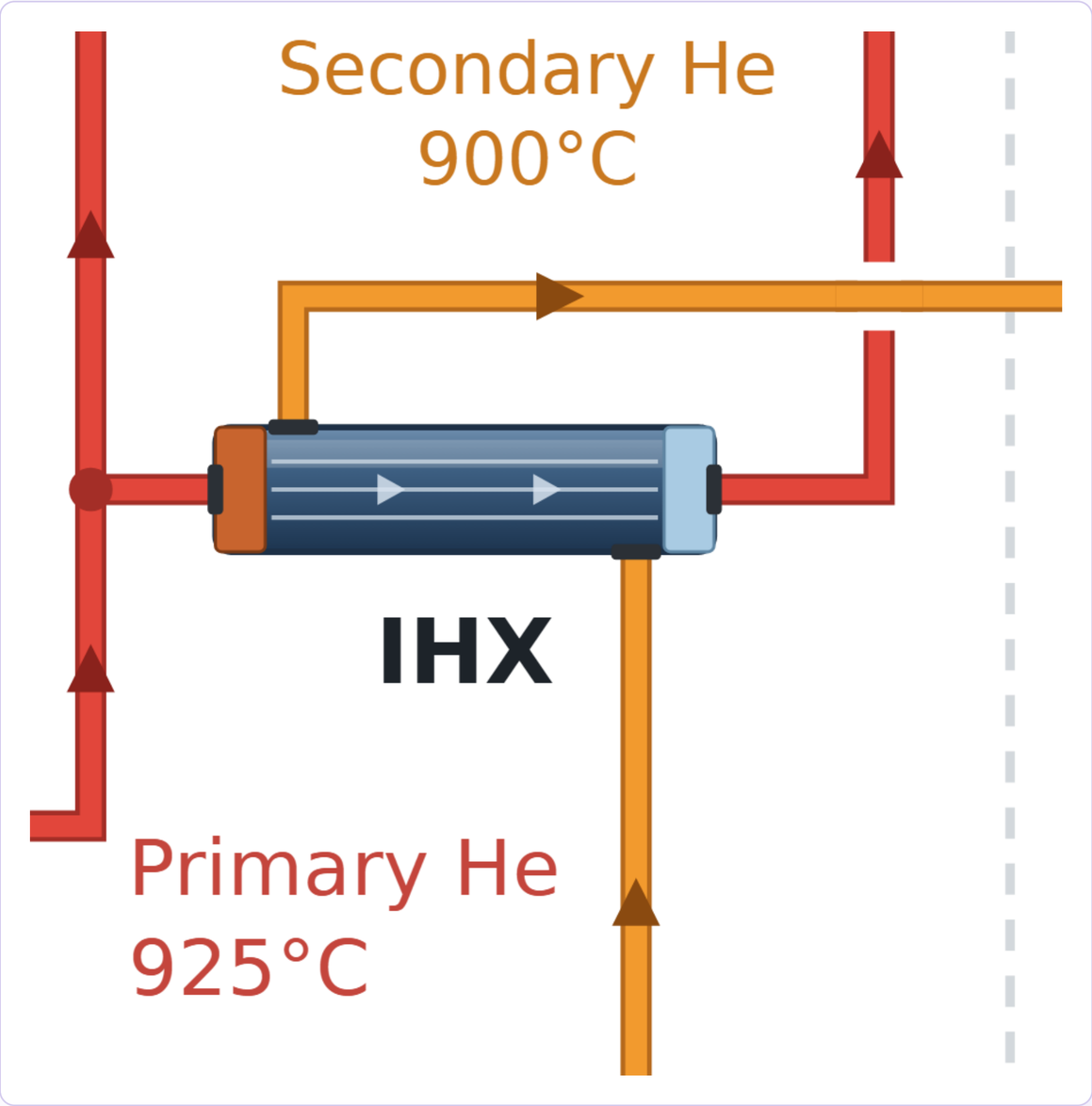
TRISO: the **first barrier**

1,600 °C

fuel limit

TRISO coatings are the first barrier to fission product release, provided fuel stays below about 1,600 °C.

Source: IAEA-TECDOC-1236 printed p14, p24, p54, p104; INL, TRISO fuel qualification



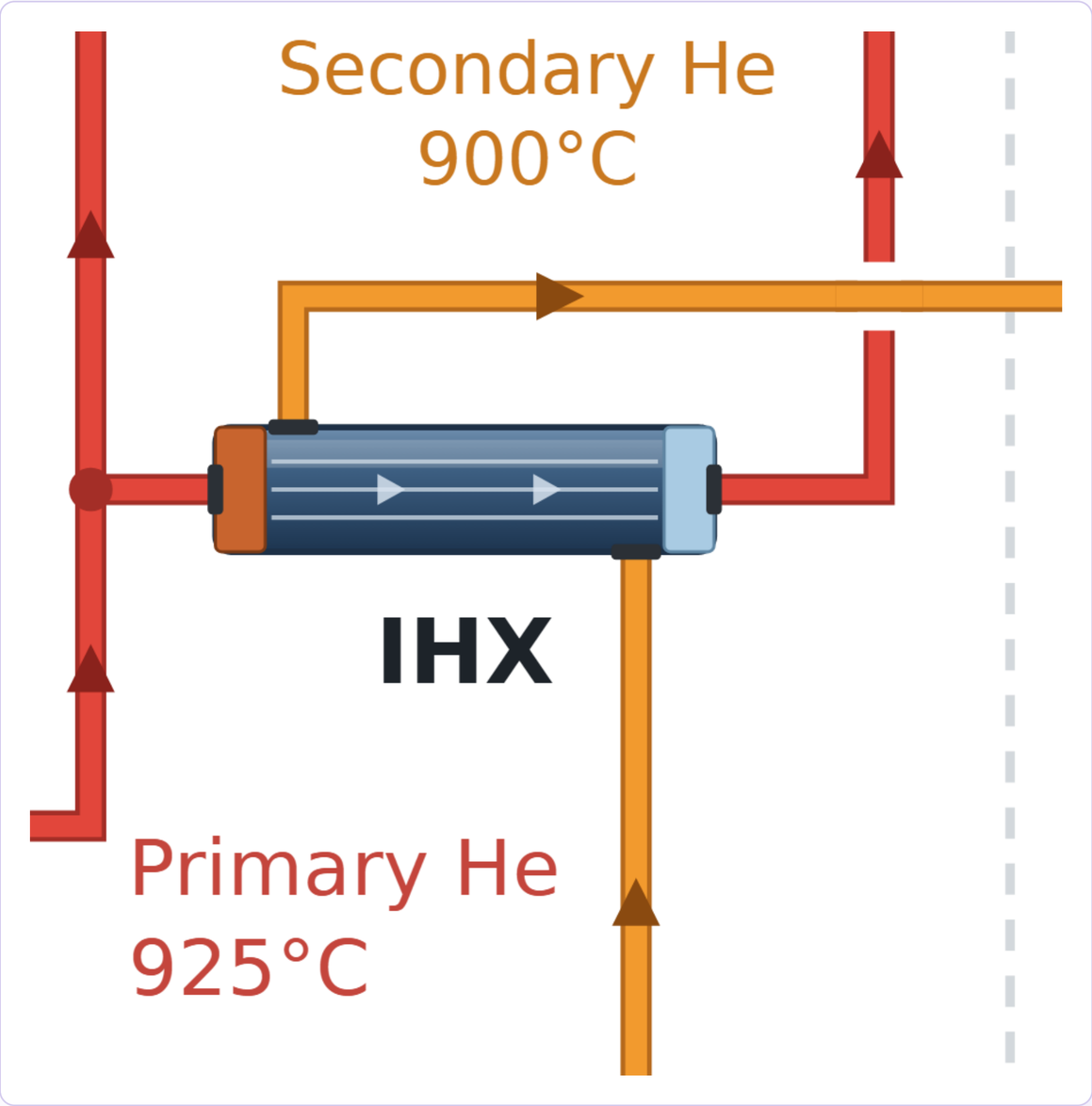
RISKS: SAFETY

The IHX: the **boundary** to isolate

First boundary

Secondary helium held slightly above primary pressure. Isolation valves are required and the figure omits them.

Source: IAEA NP-T-4.3 printed p11, p33; IAEA-TECDOC-1382 printed p4; IAEA-TECDOC-1236 printed p54 to p55, p59; Tochio and Nakagawa 2011 PDF p9; Sato et al. 2006 PDF p11



RISKS: SAFETY

900 °C helium is still unproven

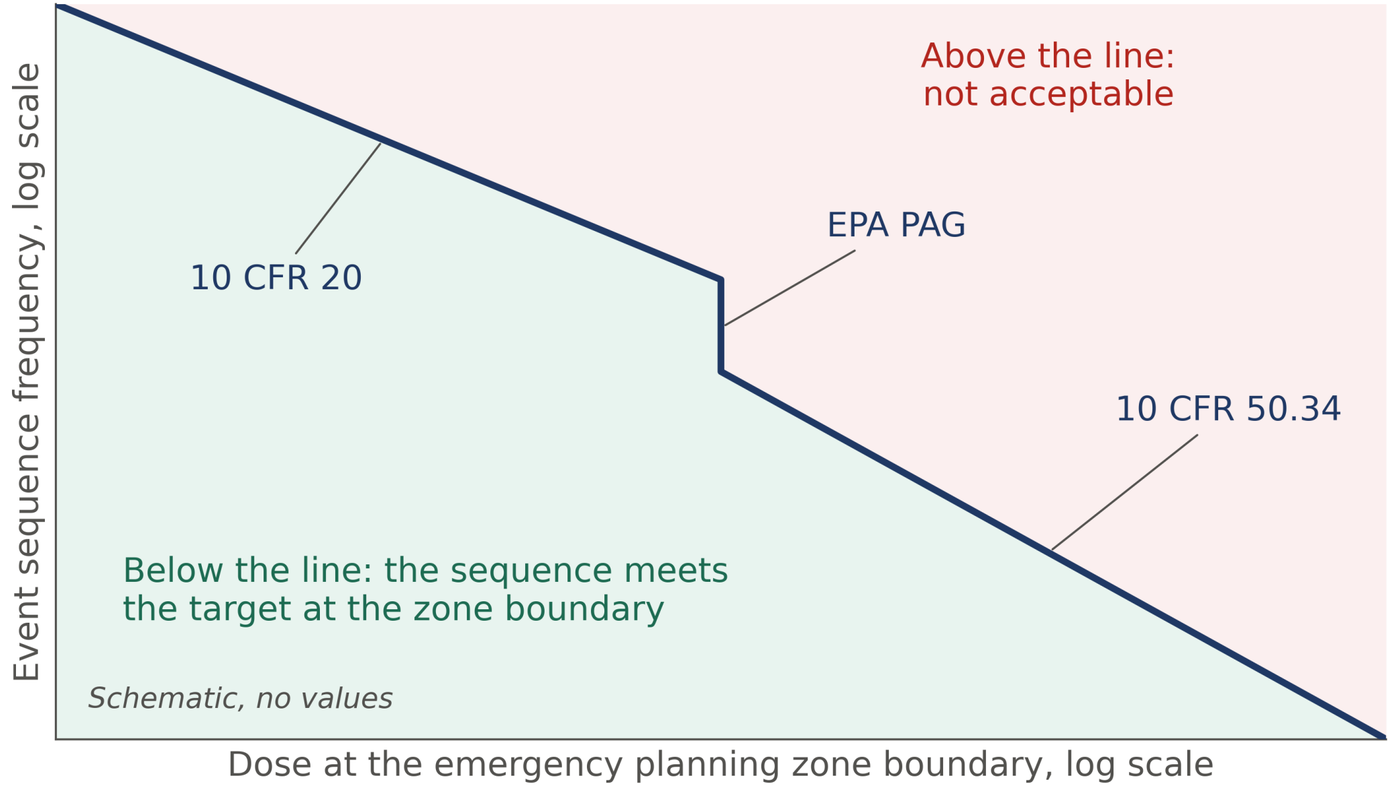
859 °C measured

900 °C needed

25 K approach is INL's assumption. At 50 K the reformer pinch falls to 4 K.

Source: IAEA NP-T-4.3 printed p14; Tochio and Nakagawa 2011 PDF p4, p8; INL TEV-961 printed p10; Sato et al. 2006 PDF p1. Status: reduced-order balances anchored on INL cases.

Emergency zone: set by a **safety case**



Schematic of the method in the Xe-100 application, with no values. The Q&A backup slides show the plotted target in rem.

No distance claimed

A low-power TRISO HTGR may justify a smaller EPZ, subject to a design-specific safety case.

Source: IAEA NP-T-4.3 printed p18, p32 to p35; SingStat; Long Mott application, Xe-100 EPZ report

May 2026

Singapore announces the IAEA INIR review

26 October 2026

METI and EMA launch manpower schemes

2027

INIR Phase 1 assessment, tentative date

RISKS: REGULATION

Singapore is **building capability**

2027

INIR Phase 1 assessment, tentative date, 19 areas

Singapore has made no decision on nuclear energy.

Source: METI, 19 May 2026; EMA and METI, 26 October 2026; Singapore Energy Lecture 2026 para 50

NOW

SNRSI public lectures

Open to anyone, including students



BEFORE ANY SITE

Public understanding

Needed ahead of any site decision

RISKS: COMMUNITY

Acceptance has to be **earned**

Public understanding comes before any site decision.

SNRSI public lectures are open to anyone, including students.

Source: SNRSI Lectures; The Straits Times, 18 May 2026; IAEA NG-G-5.1

New western island

Plans are in the works to reclaim an island for future energy, industry and defence use. The reclamation over the coming decades could merge several of Singapore's offshore islands, including Pulau Semakau, Pulau Bukom and Pulau Sudong.



Source: PRIME MINISTER'S OFFICE
PHOTO: GOOGLE EARTH
STRAITS TIMES GRAPHICS

IMPLEMENTATION LOCATION

Proposed site: **Western Island**

The author's proposal. No official approval.

Links to the island are unpriced. Land may not exist at the milestone dates.

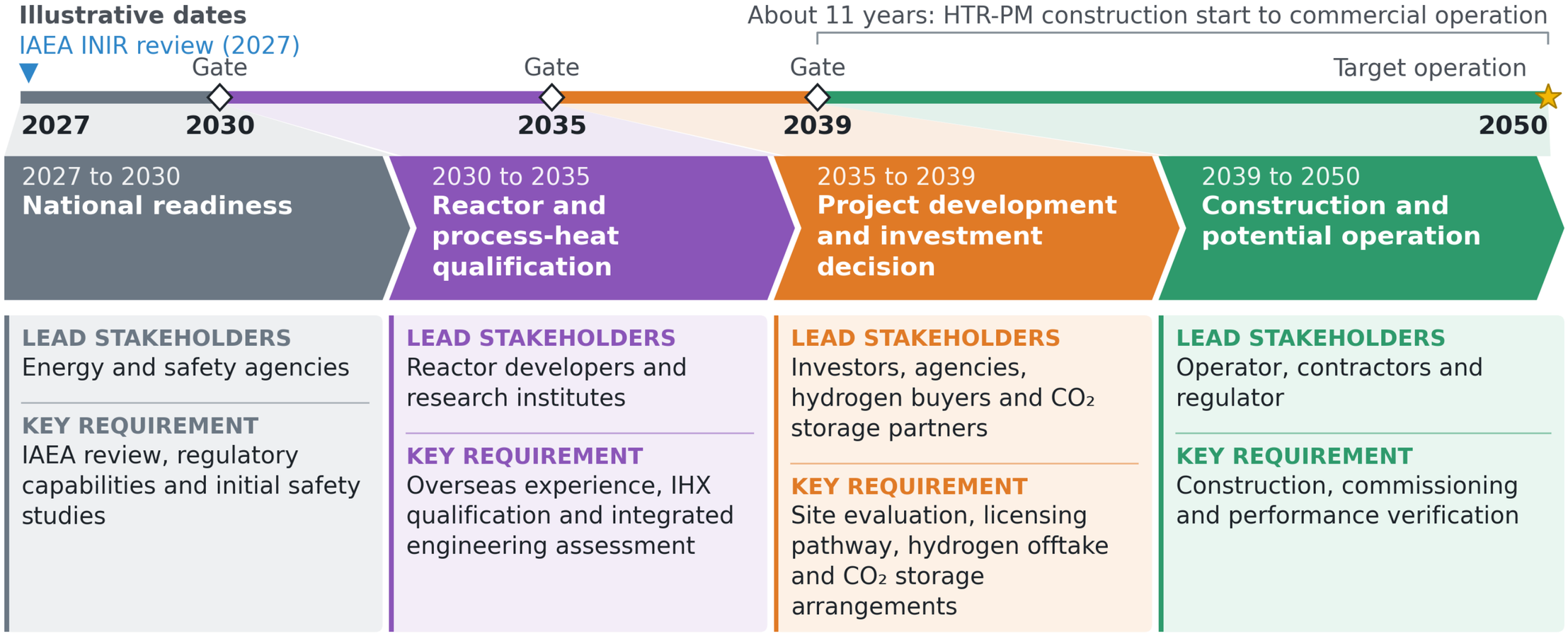
Source: gov.sg NDR 2026; BritCham Singapore 2026; Decisions D05, X15

IMPLEMENTATION ROADMAP

2027 to 2050, every date illustrative

2050
Potential operation, illustrative

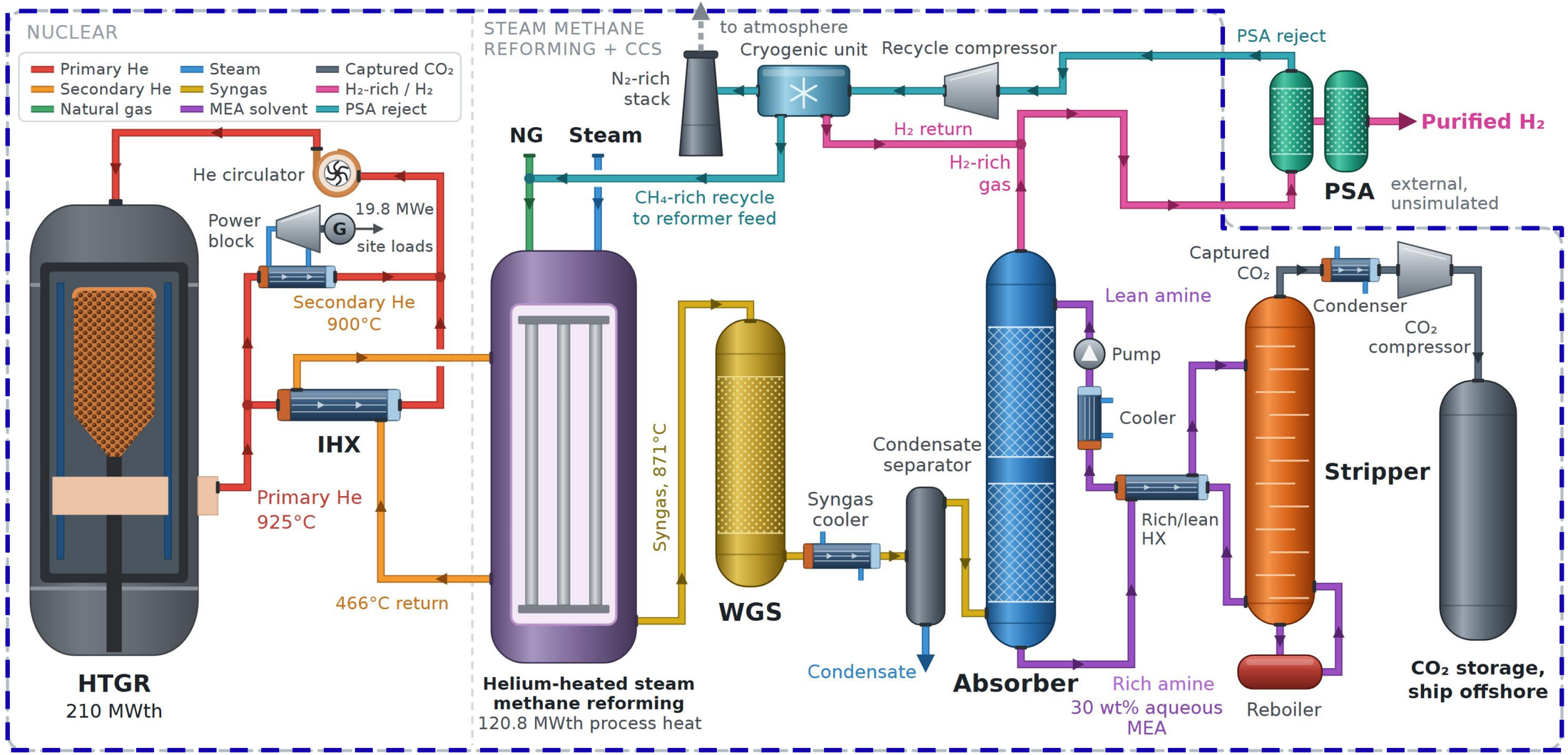
Illustrative dates.



Source: World Nuclear Association, HTR-PM; METI, 19 May 2026; Singapore Energy Lecture 2026

Where the **boundary** runs

Inside: reactor to ship offshore storage. Outside: PSA, upstream gas, construction.



Source: Decisions D06, X10; METI, 13 June 2025. Status: reduced-order balances anchored on INL cases.

$$A = E_{\text{base}} - E_{\text{atm}}$$

Net abatement equals baseline emissions less emissions to atmosphere

0.578 = 0.616 − 0.038 Mt CO_2e/y

EMISSIONS METHOD

Same hydrogen, **two plants**

616,361

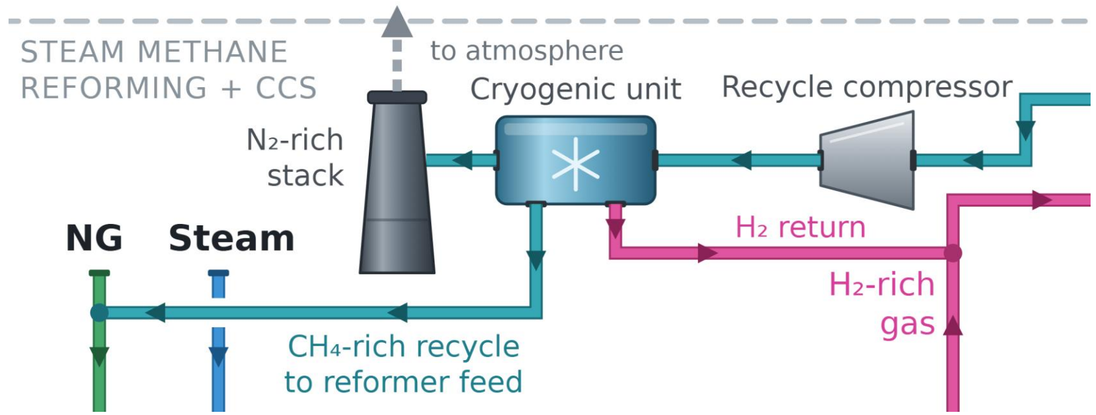
t CO_2/y baseline

9.21 kg/kg on INL's basis, bounded by 8.996 and 9.35.

Source: INL TEV-953 Table 2, PDF p30; Cho et al. 2022; IEAGHG 2017-02 Table 2; Decisions D05, D07.
Status: reduced-order balances anchored on INL cases.

EMISSIONS RESULT

What still **reaches the air**



Where the baseline carbon goes, t CO_2/y



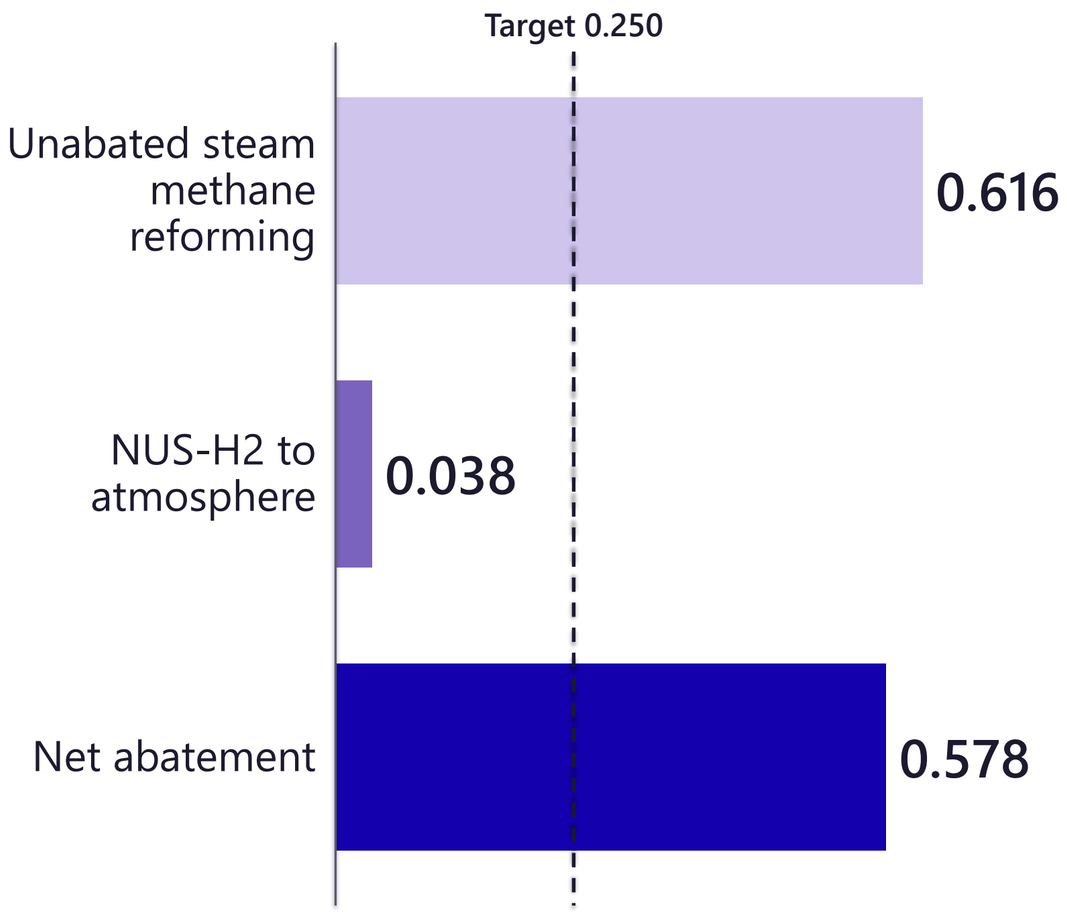
To storage | Never formed | To air **38,309**

38,309

t CO_2/y to atmosphere

0.57 kg CO_2 per kg H_2. 359,777 t/y goes to storage.

Source: Emissions_Ledger.csv; Decision D04. Status: reduced-order balances anchored on INL cases.



EMISSIONS RESULT

Net abatement **clears the target**

0.578

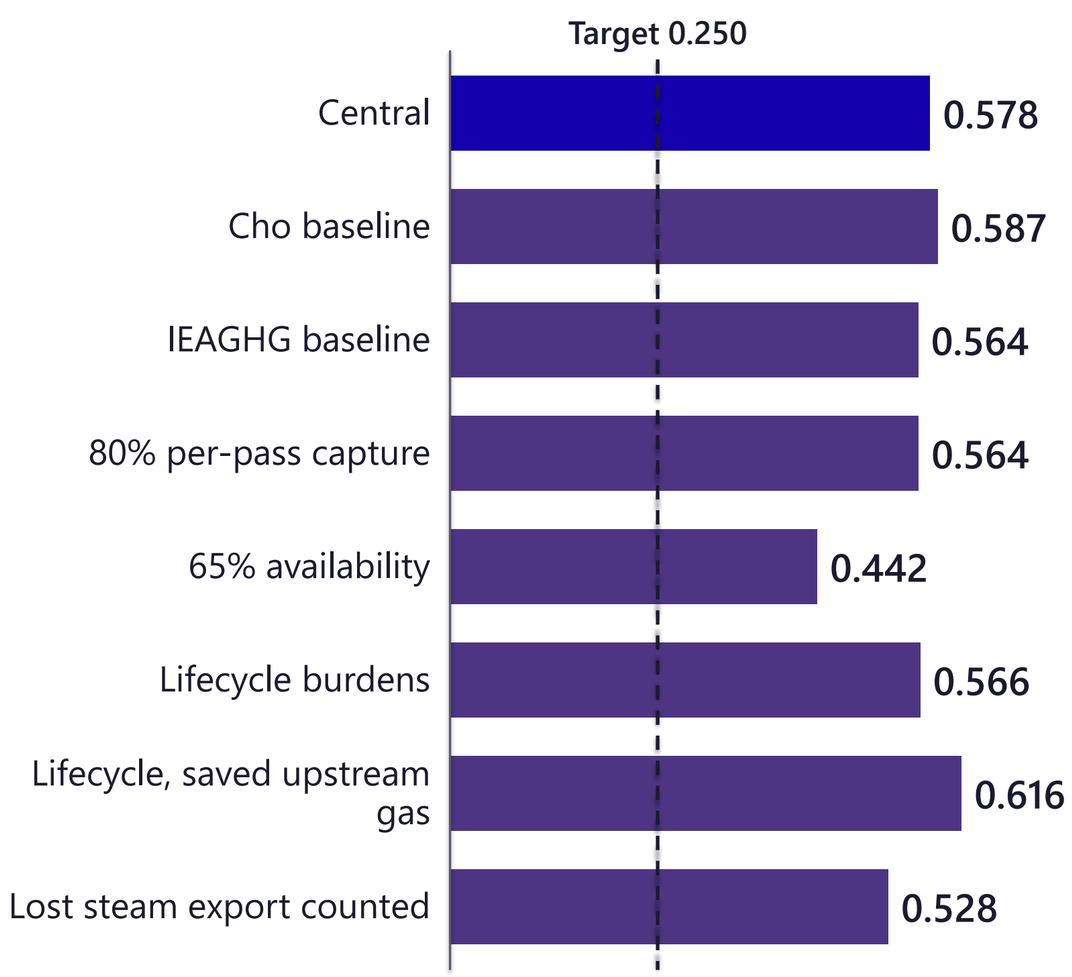
Mt CO_2e/y, 93.8% below unabated steam methane reforming

0.569 Mt/y with the storage chain burden charged. Target 0.250 Mt/y: met.

Source: Emissions_Ledger.csv; NCCS 2035 NDC; Decisions X9, X14. Status: reduced-order balances anchored on INL cases.

SENSITIVITY

Stress tests hold above target



0.442

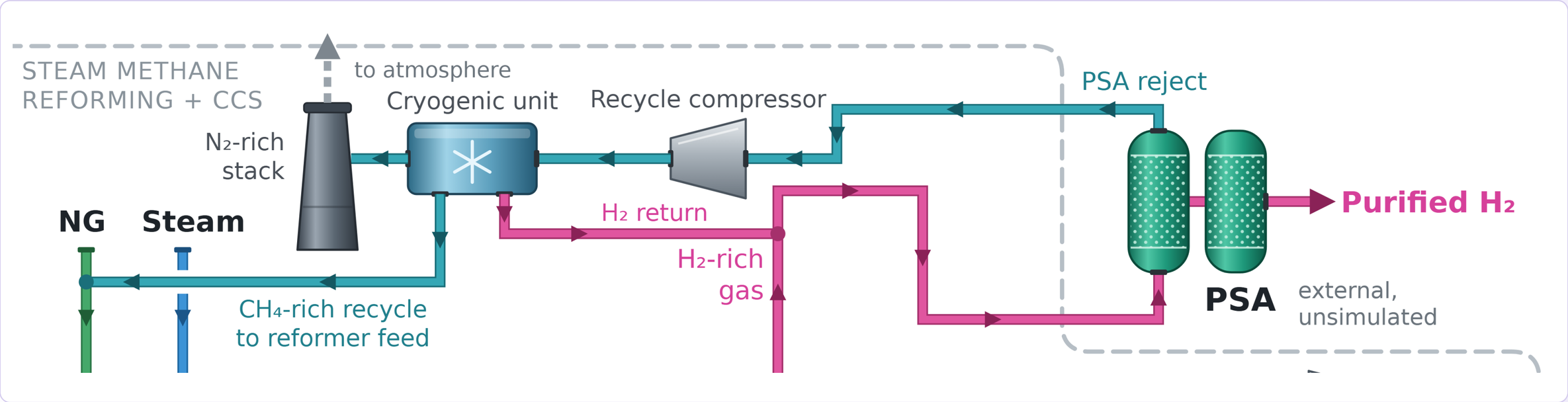
Mt/y at 65% availability

One input at a time. Corner runs carry no probability.

Source: Sensitivity_Results.csv; Emissions_Ledger.csv; IAEA NP-T-4.3 printed p38. Status: reduced-order balances anchored on INL cases.

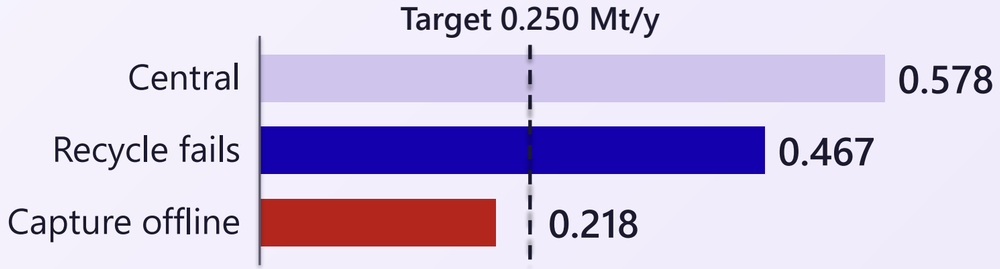
SENSITIVITY

The assumption that **matters most**



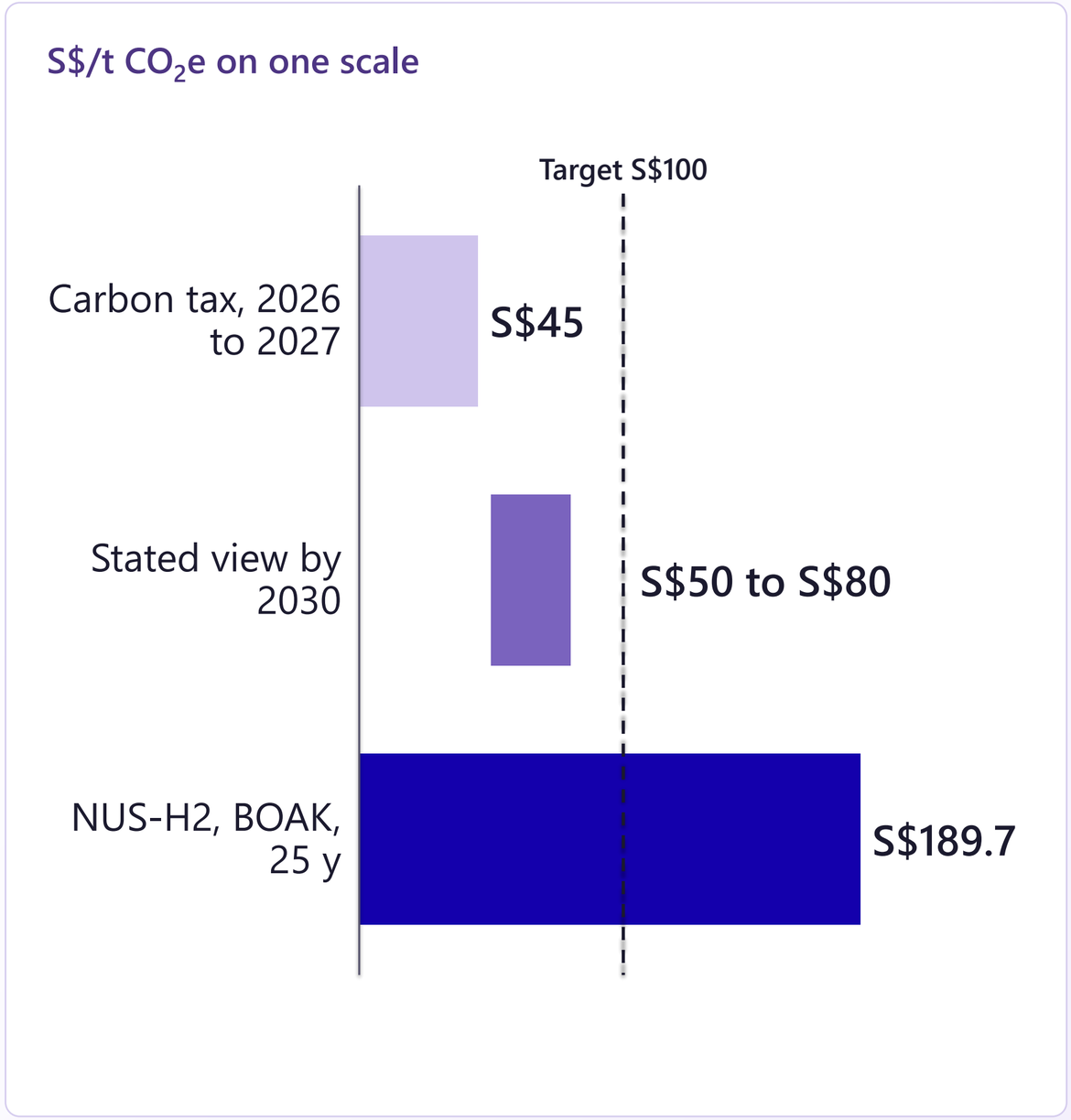
0.467

Mt/y if recycle fails



Gas rises to 30.4 MMSCFD. With capture offline, only 0.218 Mt/y remains.

Source: Decision X20; Sensitivity_Results.csv tail gas rows. Status: reduced-order balances anchored on INL cases.



ECONOMICS

What the **market** compares NUS-H2 with

S$45/t

carbon tax, 2026 to 2027

A stated view of S$50 to S$80/t by 2030. No offtake is contracted.

Source: NCCS Carbon Tax; Air Liquide and Aster 2025. Status: reduced-order balances anchored on INL cases.

$$AC = \frac{C_{proposed} - C_{baseline}}{E_{baseline} - E_{proposed}} = \frac{\Delta C_{annual}}{\Delta E_{net}}$$

AC, abatement cost

Annual ledger

Each cost difference against unabated steam methane reforming, levels L0 to L5

Discounted cash flow

Both plants, year by year, at equal hydrogen output

ECONOMICS METHOD

Cost, **rebuilt two ways**

0.47%

largest gap between the methods, over all 72 points

0.2% on the headline grid. Equal hydrogen output, 8% real discount rate. Shared inputs, so agreement tests the arithmetic.

Source: Decision D08; Economic_Ledger.csv. Status: reduced-order balances anchored on INL cases.

COST ASSUMPTIONS

The inputs that **drive cost**

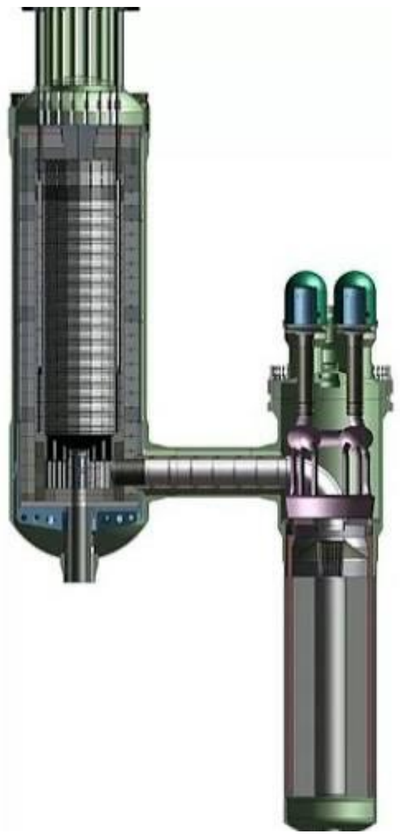
S$15/GJ
Gas, the largest credit

Input	Low	Central	High	Unit
Gas	10	15	20	S$/GJ
Transport and storage	23.2	60	112	S$/t
Reactor capital	1,456	2,184	2,548	US$/kWt

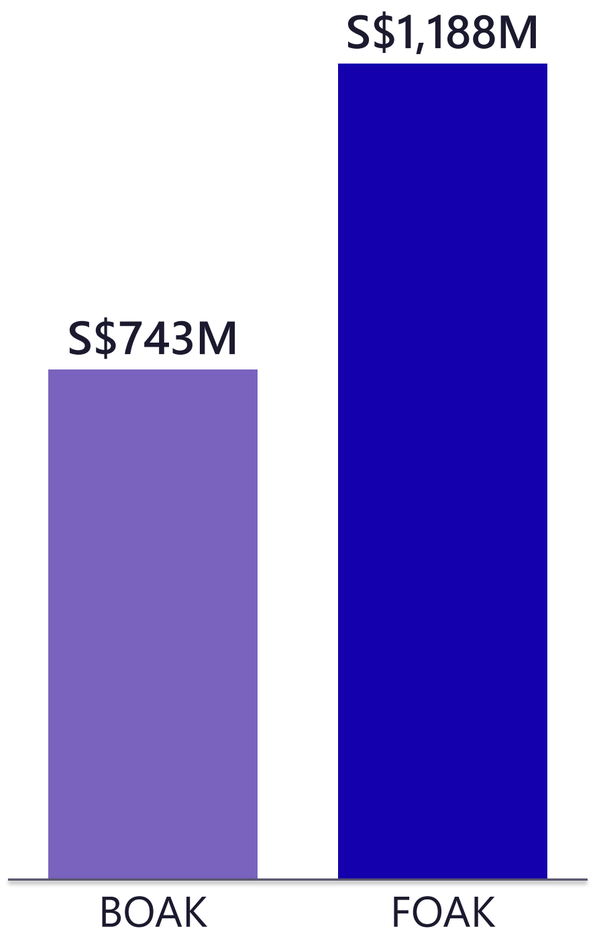
Source: INL/RPT-23-72972 PDF p47; IEAGHG 2017-02 Tables 2, 3; IMF via FRED, PNGASJPUSDM; ERIA 2024; METI PQ, 8 April 2025; Decision D10, X18, X19

REACTOR COST

Reactor capital: **first plant** against later plants



Analogue only



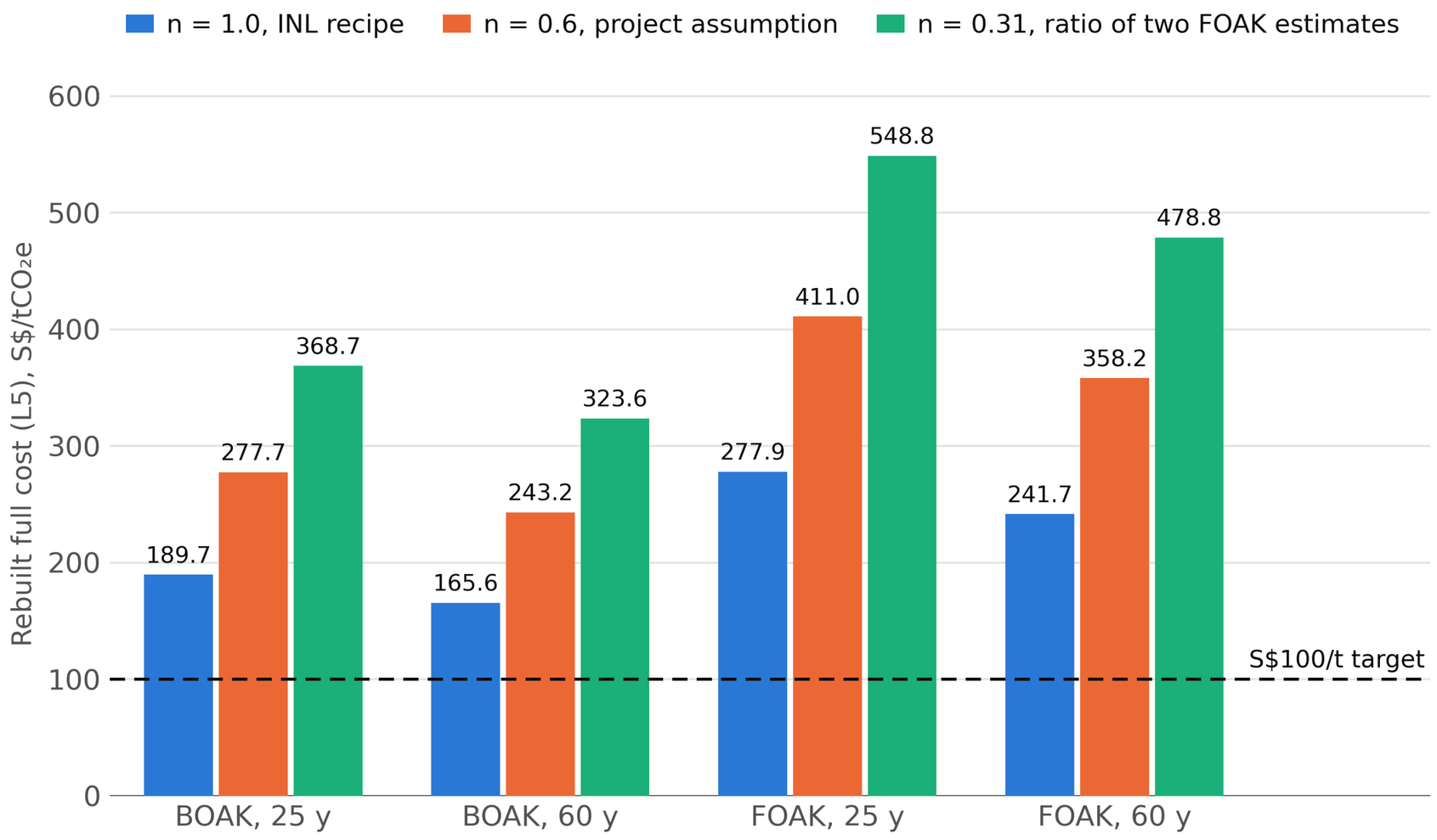
S$743M BOAK

S$1,188M FOAK

Generic HTGR. Xe-100 is an analogue only, with no vendor quotation.

Source: INL/RPT-23-72972 Table 1, PDF p5, p47; X-energy; Decisions D09, D23, X4. Status: reduced-order balances anchored on INL cases.

The **scale law** moves everything



S$189.7

n = 1.0, INL recipe

S$277.7

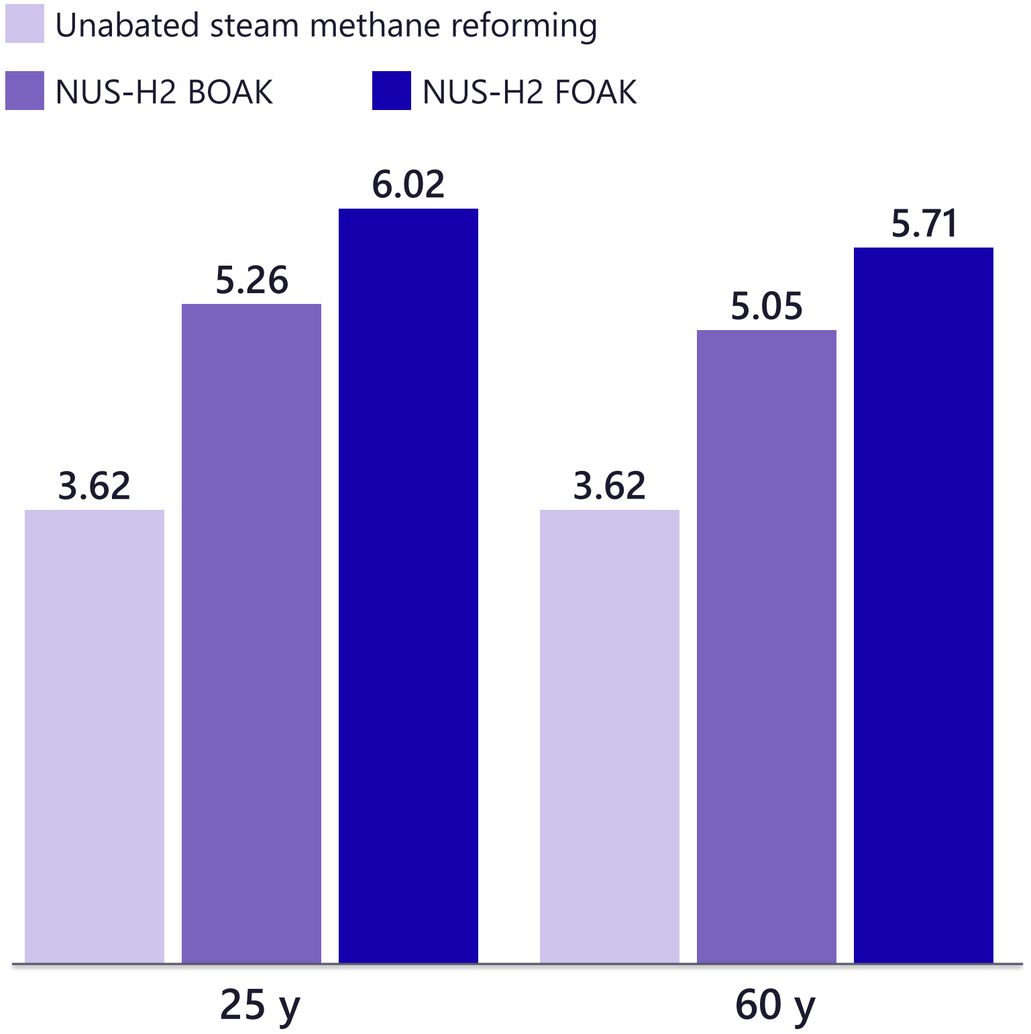
n = 0.6, project assumption

S$368.7

n = 0.31, ratio of two FOAK estimates

per tonne at BOAK, 25 y

Source: INL/RPT-23-72972 PDF p25, p27, p39, p45; Decisions D09, X5. Status: reduced-order balances anchored on INL cases.



Levelised cost of hydrogen, S$/kg

HYDROGEN COST

Hydrogen cost, **both plants**

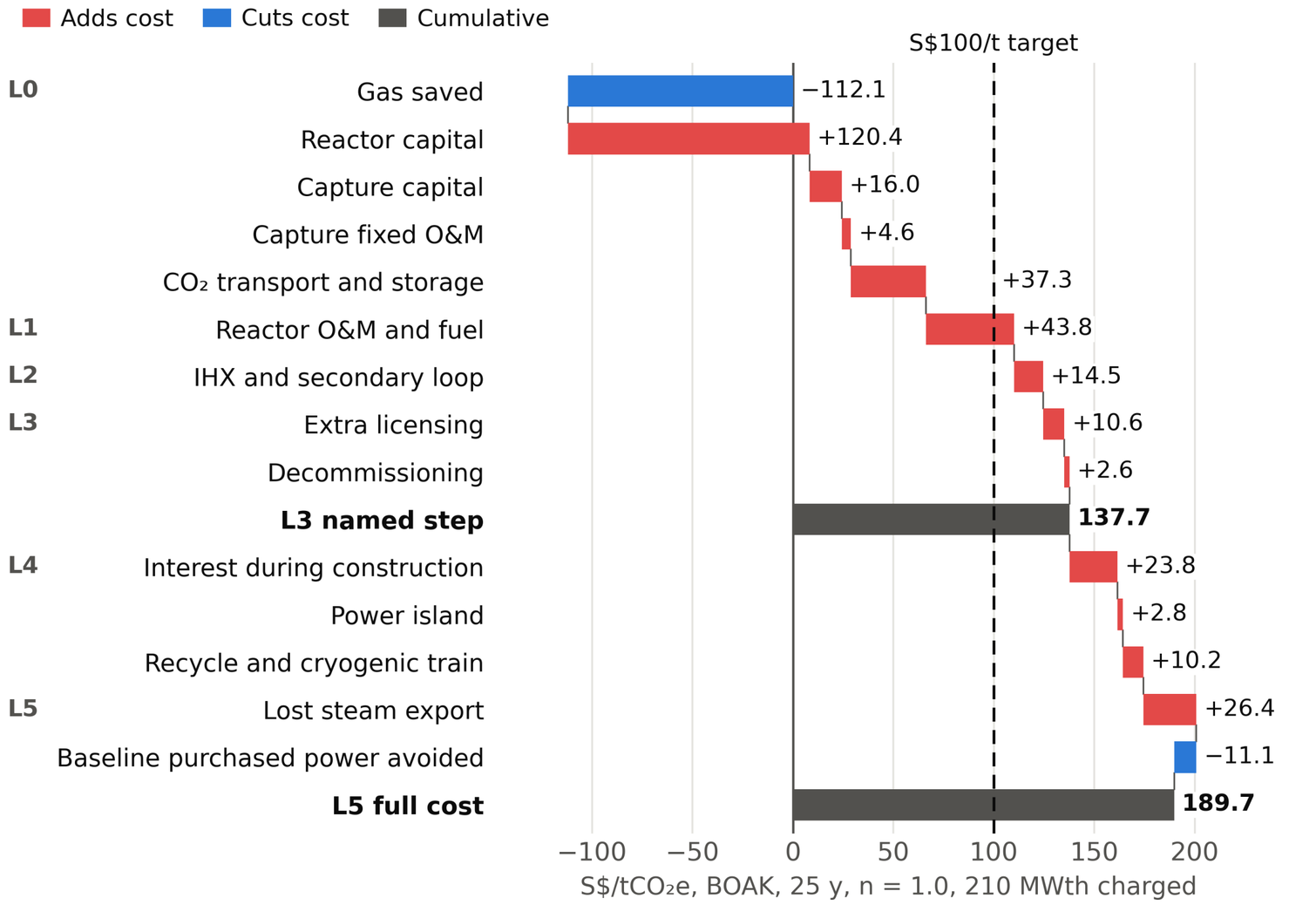
S$3.62/kg unabated steam methane reforming

S$5.26/kg NUS-H2, BOAK, 25 y

FOAK S$6.02/kg. At 60 y: BOAK S$5.05/kg, FOAK S$5.71/kg.

Source: Economic_Ledger.csv cash flow rows; IEAGHG 2017-02 Tables 3, 4, PDF p17, p18. Status: reduced-order balances anchored on INL cases.

Every real cost is **on the ladder**



S$189.7/t

Rebuilt full cost

BOAK, 25 y, n = 1.0, 210 MWth charged. Named step L3: S$137.7/t.

Source: Economic_Ledger.csv; Decision D08. Status: reduced-order balances anchored on INL cases.

ABATEMENT COST

Levelised abatement cost: target missed

Levelised abatement cost, discounted cash flow, S$/t

BOAK, 25 y

S$190.1

Ledger S$189.7

BOAK, 60 y

S$166.0

Ledger S$165.6

FOAK, 25 y

S$278.4

Ledger S$277.9

FOAK, 60 y

S$242.2

Ledger S$241.7

0 of 8

central cases reach S$100/t

Source: Economic_Ledger.csv; Decisions D08, X4. Status: reduced-order balances anchored on INL cases.

ABATEMENT COST

What S$100/t would take

US$824/kWt

Needed, against INL's low of US$1,456/kWt

Lever	Needed at 25 y	Needed at 60 y	Tested range
Gas, S$/GJ	30.7	26.5	To 20
Reactor, US$/kWt	824	1,031	From 1,456
T&S, S$/t	−84.2	−45.5	From 23.2
Sized to need, S$/t reached	161.2	140.4	Still above target

Source: Report section 4.7. Status: reduced-order balances anchored on INL cases.

ABATEMENT COST

S$96/t: S$100/t target met for a 10th-of-a-kind plant over 60 years with gas at S$20.7/GJ

MATURE_60Y: 10th-of-a-kind on INL's BOAK learning route, 60 y, gas at S$20.7/GJ

S$95.6/t
Annual ledger, L5

S$95.9/t
Discounted cash flow

S$20.7/GJ
Gas price that meets the target

S$15/GJ
Central gas price

Higher gas prices lower the abatement cost, because NUS-H2 burns less gas than an unabated steam methane reforming plant of the same output. S$20.7/GJ lies above the S$10 to S$20/GJ range tested. At S$20/GJ the scenario gives S$99.6/t on L5 and S$99.9/t by DCF.

Base full cost

BOAK, 25 y, n = 1.0, 210 MWth charged

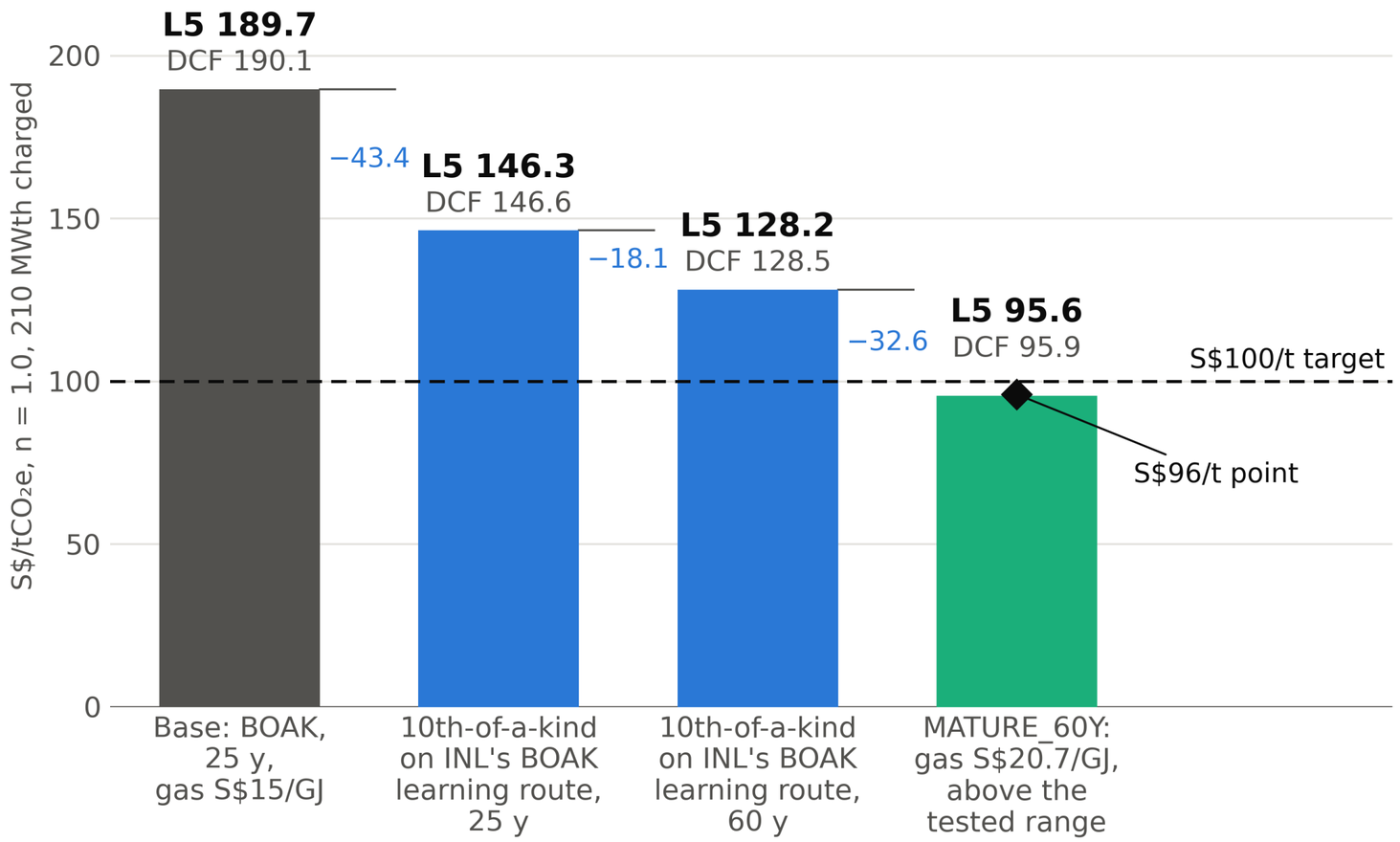
S$189.7/t
Annual ledger, L5

S$190.1/t
Discounted cash flow

S$100/t target missed at the base

Source: Sensitivity_Results.csv MATURE_60Y rows; Economic_Ledger.csv; INL/RPT-23-72972 PDF p46. Status: reduced-order balances anchored on INL cases.

How the cost **steps down** to the target



Base

S$189.7/t

DCF S$190.1/t

After the three steps

S$95.6/t

DCF S$95.9/t

The reactor steps cut S$43.4/t for a 10th-of-a-kind unit and S$18.1/t for a 60 year life. Gas at S$20.7/GJ then cuts S$32.6/t.

Source: Sensitivity_Results.csv STEPDOWN rows; INL/RPT-23-72972 PDF p46. Status: reduced-order balances anchored on INL cases.

Overall **feasibility**

Carbon

0.578 Mt/y

Abatement met

On this boundary, with working capture and recycle

Cost

S$165.6 to S$277.9/t

Cost missed

At central inputs. Met at S$96/t for a 10th-of-a-kind plant on INL's BOAK learning route over 60 y with gas at S$20.7/GJ, above the tested range

Technology

IHX

Unqualified

No IHX built at this duty

Site and policy

Open

No decision

Land and a national decision do not yet exist

Source: Report sections 4 and 5. Status: reduced-order balances anchored on INL cases.

LIMITS OF THE MODEL

What this model has **yet to show**

Still to close

- Recycle transfer to MEA
- Ship liquefaction load
- Capture capital basis
- SRD on syngas
- HTGR availability
- Site links
- Backup firing capital
- Offshore accounting

360 of 650

checks are independent of the source data

325 pass (92 on independent data), 32 fail, 251 reported without a limit, 42 not run.

Source: Validation_Results.csv; Design decisions sections 4 and 9. Status: reduced-order balances anchored on INL cases.

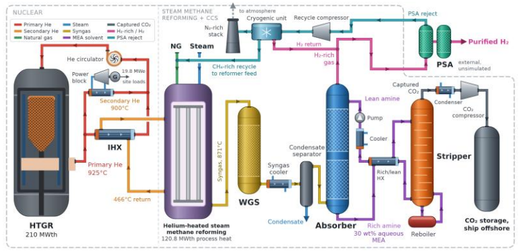
The author

Wee Zhern Ray

CONCLUSION

NUS-H2 in one slide

What



210 MWth

Helium-heated steam methane reforming with MEA capture

Carbon

0.578 Mt/y

Target met

93.8% cut against same-size unabated steam methane reforming

Cost

S$189.7/t

Base: target missed

BOAK, 25 y. Target met at S$96/t for a 10th-of-a-kind plant on INL's BOAK learning route over 60 y with gas at S$20.7/GJ, above the tested range

Next

2050

IHX qualification and a design-specific safety case, on an illustrative path

Source: Emissions_Ledger.csv; Economic_Ledger.csv; Sensitivity_Results.csv MATURE_60Y rows. Status: reduced-order balances anchored on INL cases.

Reference Frame: variables

Symbol	Definition	Units	Central
$\dot{m}_{H_2}$	Annual hydrogen	t/y	66,925
a	Availability	%	85
T_{ROT}	Reactor outlet	°C	925
T_{He}	Secondary helium supply	°C	900
R_{rated}	Reactor rating	MWth	210
R_{need}	Reactor needed	MWth	178.5
Q_{proc}	Process heat	MWth	120.8
E_{site}	Site electrical load	MWe	19.8
e_{base}	Baseline intensity	kg CO_2/kg H_2	9.21

Symbol	Definition	Units	Central
E_{base}	Baseline emissions	t/y	616,361
E_{atm}	Emissions to atmosphere	t/y	38,309
A	Net abatement	Mt/y	0.578
η_{pass}	Per-pass capture	%	90
η_{plant}	Plant capture	%	90.4
q_{reb}	Specific reboiler duty	GJ/t	3.8
r	Real discount rate	%	8
n	Reactor size exponent	none	1.0
L	Learning rate per doubling	%	10

Source: Final_Design_Basis.csv. Status: reduced-order balances anchored on INL cases.

Reference Frame: equations

(1) $A = E_{\text{base}} - E_{\text{atm}}$

(2) $E_{\text{base}} = e_{\text{base}}\,\dot{m}_{H_2}$

(3) $\eta_{\text{plant}} = m_{\text{cap}} \,/\, (m_{\text{cap}} + E_{\text{atm}})$

(4) $R_{\text{need}} = Q_{\text{IHX}} + Q_{\text{loss}} + Q_{\text{power}} - W_{\text{shaft}}$

(5) $Q = \dot{m}\, c_p\, \Delta T$

(6) $\text{CRF} = r \,/\, (1 - (1 + r)^{-N})$

(7) $\text{AC} = \Delta C \,/\, A$

(8) $\text{LCOH}_x = \text{NPV}(c_x) \,/\, \text{NPV}(m_{H_2})$

(9) $\text{LAC} = \text{NPV}(\Delta c) \,/\, \text{NPV}(A)$

(10) $C(S) = C_{\text{ref}}\,(S \,/\, S_{\text{ref}})^n$

(11) $C_N = C_1\,(1 - L)^{\log_2 N}$

Ranges: SRD 3.4 to 4.2 GJ/t, life 25 to 60 y, 1.3 to 2.1 FOAK premium. CRF at 25 y is 0.0937 and at 60 y is 0.0808.

Source: Project derivations; INL/RPT-23-72972 PDF p46, p47. Status: reduced-order balances anchored on INL cases.

BACKUP

Steam methane reforming plus CCS comparator

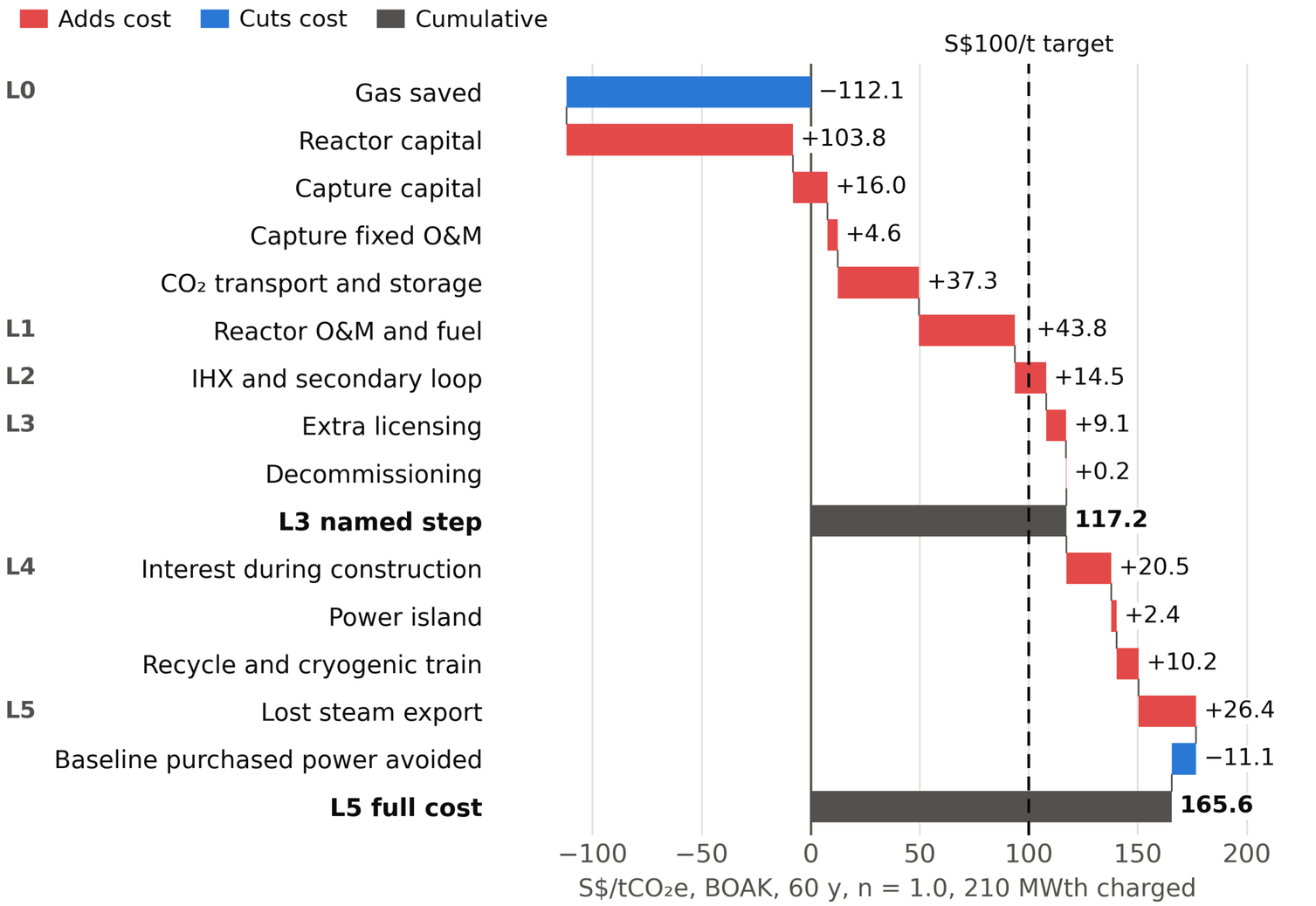
S$203.0/t
IEAGHG Case 3 at matched T&S

Item	NUS-H2	IEAGHG Case 3
Capture route	MEA on shifted syngas	MEA on flue gas
Overall capture	90.4%	90.0%
Baseline, kg/kg	9.21	8.996
Abatement, Mt/y	0.578	0.536
Cost as published, S$/t	not applicable	162.1
Cost at T&S S$60/t, S$/t	BOAK 189.7, FOAK 277.9	203.0

Indicative. Gas price and availability differ from this ledger.

Source: IEAGHG 2017-02 Tables 2 to 5; Decisions D11, X17. Status: reduced-order balances anchored on INL cases.

Cost ladder at 60 y



S$165.6/t

Rebuilt full cost

BOAK, 60 y, n = 1.0, 210 MWth charged.

Source: Economic_Ledger.csv. Status: reduced-order balances anchored on INL cases.

BACKUP

Cost ladder, every level

S$57.8M/y
Ceiling at the cost target

S$/t	BOAK 25 y	BOAK 60 y	FOAK 25 y	FOAK 60 y
L0 gas, capital, capture, T&S	66.2	49.7	138.5	112.0
L1 reactor O&M and fuel	110.0	93.5	182.3	155.8
L2 IHX and loop	124.5	108.0	196.7	170.3
L3 licensing, decommissioning	137.7	117.2	209.9	179.5
L4 interest, power island, recycle train	174.5	150.4	262.7	226.4
L5 lost steam, avoided purchased power	189.7	165.6	277.9	241.7
DCF levelised abatement cost	190.1	166.0	278.4	242.2

Source: Economic_Ledger.csv. Status: reduced-order balances anchored on INL cases.

Assumptions: plant

Item	Value	Status
Reactor outlet, IHX approach	925 °C, 25 K	INL assumption, 50 K tested
Secondary helium	900 °C out, 466 °C back	INL
Reformer outlet	871 °C	INL equilibrium assumption
Power conversion	39.7%	INL, 47.5% tested
MEA	30 wt%, rich 0.50, lean 0.25	Project assumption
Absorber, stripper	26.8 bar, 2 bar	INL, amine specification
SRD	3.8 GJ/t, 3.4 to 4.2	Assumed from flue gas pilots
Availability	85%	Assumed, no measured HTGR record

Source: Final_Design_Basis.csv; INL TEV-961; Decisions D26 to D33

Assumptions: cost

Input	Low	Central	High	Unit
Reactor overnight capital	1,456	2,184	2,548	US$2019/kWt
FOAK premium	1.3	1.6	2.1	× BOAK
Reactor O&M and fuel	6	10	14	US$2019/MWth-h
Gas	10	15	20	S$/GJ
T&S	23.2	60	112	S$/t
Extra licensing	0	50	137.5	US$M
Decommissioning	750	1,000	1,250	US$2022/kWe

Discount 8%. FX S$1.3065/US$. Construction 5.5 y.

Source: INL/RPT-23-72972 Table 1; IEAGHG 2017-02; FRED, AEXSIUS; IMF via FRED, PNGASJPUSDM; ERIA 2024; Decision D10

Cost sensitivities

Change	Full cost, S$/t	Change	Full cost, S$/t
T&S S$23.2	166.8	T&S S$112	222.1
Gas S$10/GJ	218.3	Gas S$20/GJ	161.2
Availability 65%	253.0	Availability 90%	178.3
Reactor capital low	141.7	Reactor capital high	213.8
Gas saving on Ahn and Lee, carbon held	242.0	Capture capital on MEA flue gas	220.5
Sized to need	161.2	No recycle	283.1

Source: Sensitivity_Results.csv; Report section 4.5. Status: reduced-order balances anchored on INL cases.

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What are some of the notable light-water small modular reactor designs?

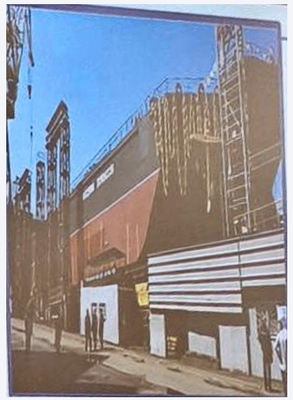
Q&A



ACP100 (CNNC)

ACP100 (China, 125 MWe)

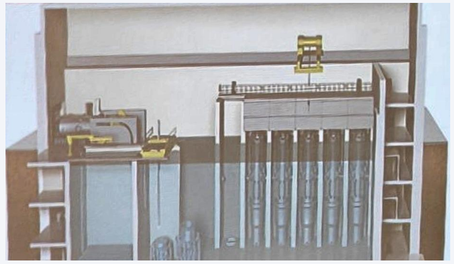
- Construction began in July 2021 at Changjiang NPP in Hainan province.
- Completed cold functional tests in 2025 and expected to be in commercial ops in 2026 or early 2027.



Akademik Lomonosov docked

KLT-40S (Russia, 2 x 35 MWe)

- Towed and reached final destination in Sep 2019.
- Connected to the grid in Dec. 2019, commercial operation at the end of May 2020.
- Expected to be one-of-a-kind.

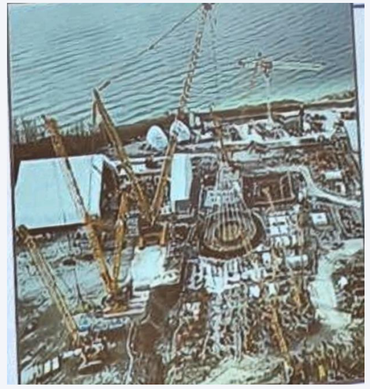


Nu*Scale* reactor modules

NuScale (USA, 12 x 77 MWe)

- Received Standard Design Approval issued by U.S.NRC in Sept. 2020.
- NRC agreed that the **small modular reactor does not need backup power** and may have a **smaller emergency planning zone**.
- UAMPS will build its first plant at Idaho National Lab.

Cancelled in Nov 2023



BWRX-300 Reactor (Installation of basemat)

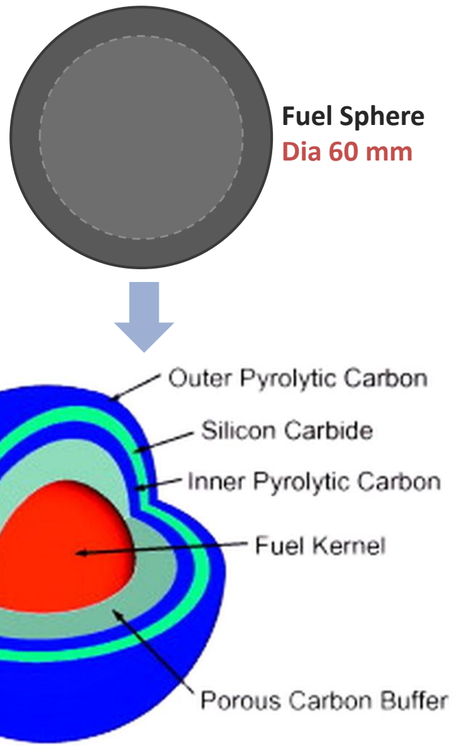
BWRX-300 (USA-Japan, 300 MWe)

- BWR developed by GE Vernova Hitachi Nuclear Energy (GVH).
- First unit construction started at Darlington, Canada in 2026.
- Also strong interest in Poland, US, etc.

What are High-Temperature Gas-Cooled Reactors?

- **High temperature gas-cooled reactors (HTGRs)** are advanced reactors that used **helium**, an inert gas, instead of water as coolant.
- Allows **higher temperature** (~750°C – 1000°C)
  - higher efficiency in electricity production
  - allows other applications, e.g., hydrogen production, petrochemical processing, etc.
- Uses **TRISO fuel** – a vastly improved fuel – in spherical balls or prismatic blocks
- The fuel is embedded into 100,000s of billiard ball-sized “pebbles” or prismatic blocks of graphite.
- Nuclear-grade graphite is used *as* neutron moderator (reduce speed of neutrons), structural materials as *well as reflector of* neutrons.

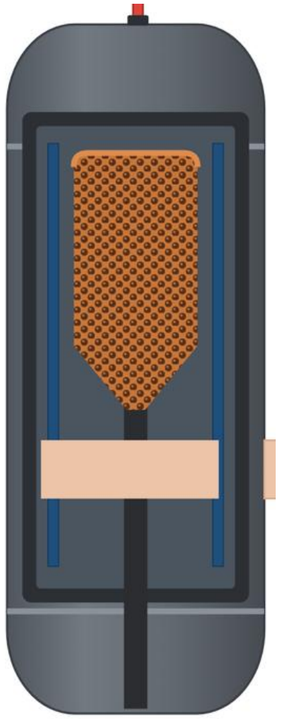
~10,000 coated particles embedded in graphite sphere



TRISO Coated Particle: Dia 0.92 mm

Fuel Kernel: Dia 0.5 mm, ~0.0007 g of UO_2

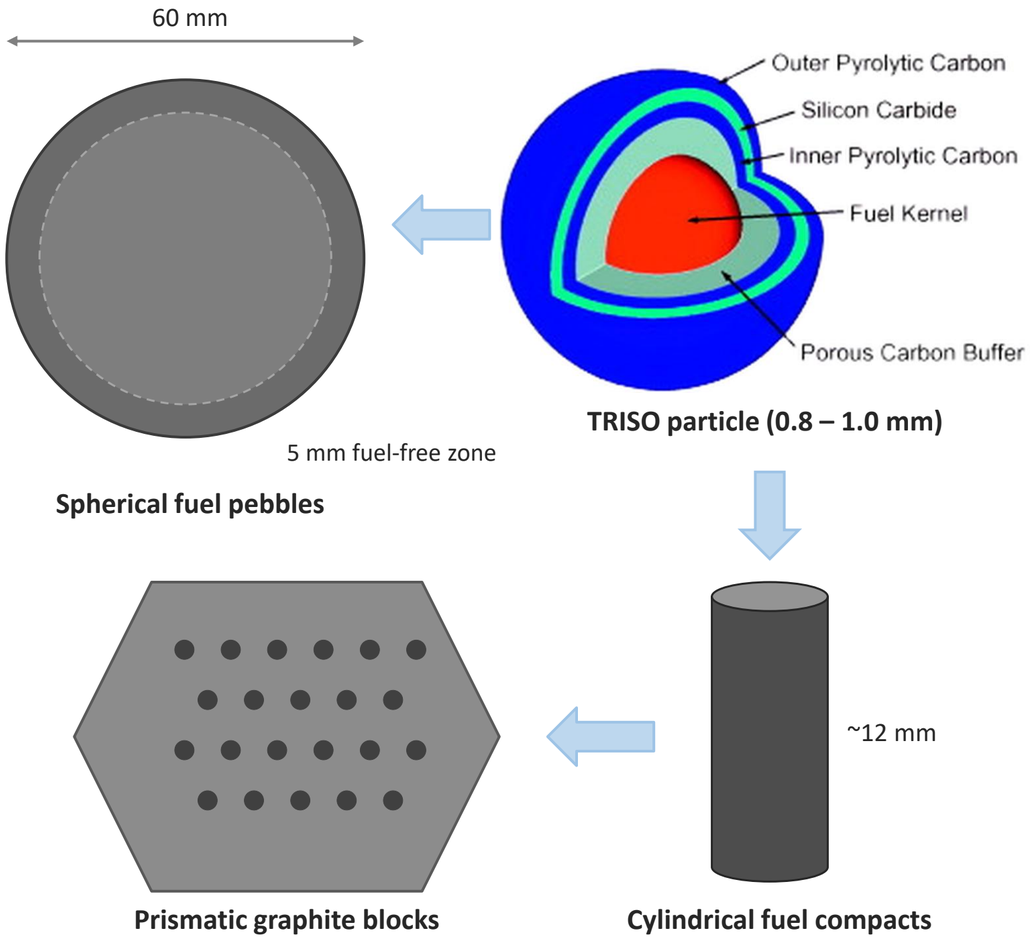
Silicon carbide barrier coating: withstand up to 1,600°C



Fuel spheres form a graphite pebble bed in reactor vessel

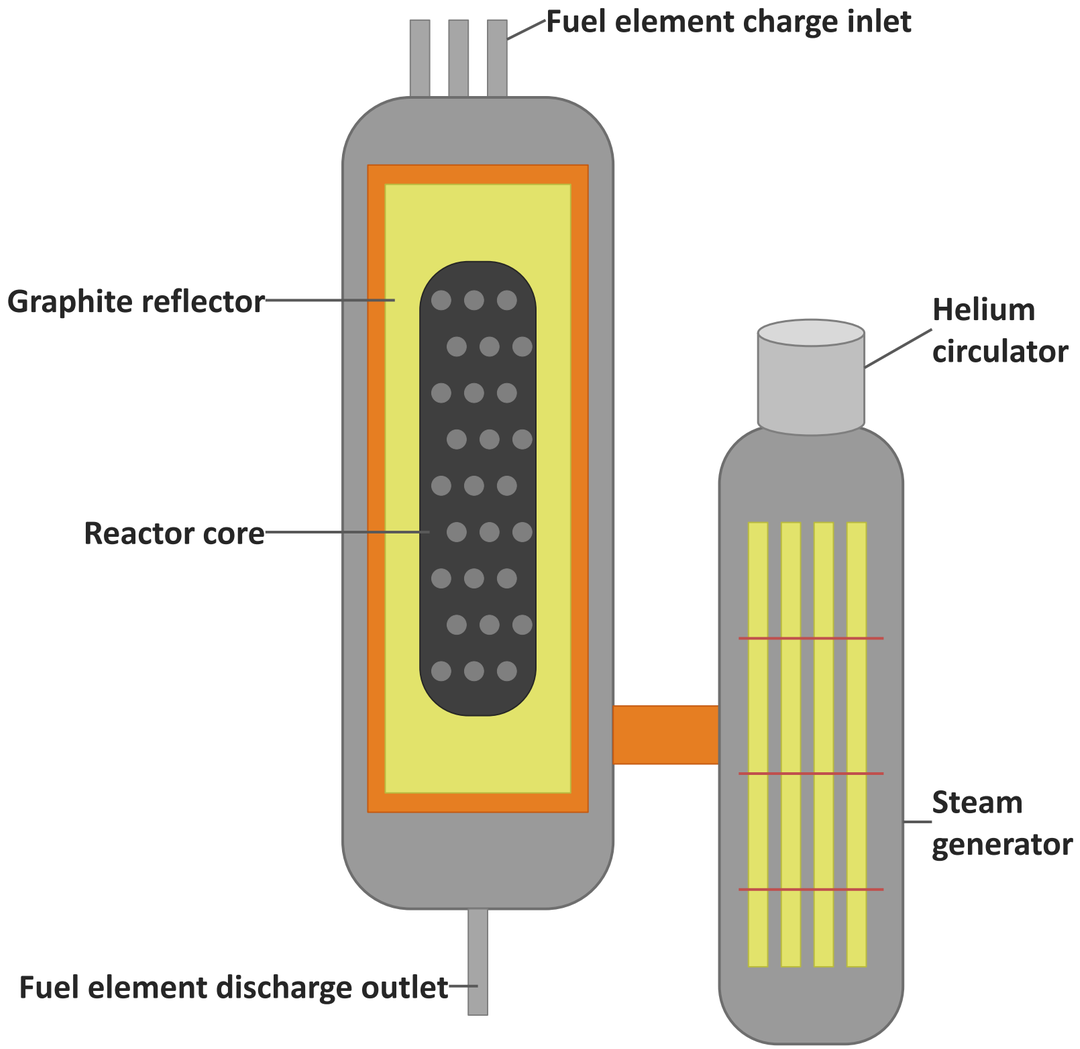
What are TRISO particles?

- Each TRISO particle consists of a kernel of uranium dioxide (UO_2) fuel (sometimes UC or UCO), which has been coated with four layers:
  - porous buffer layer made of carbon
  - dense inner layer of pyrolytic carbon (IPyC)
  - ceramic layer of silicon carbide (SiC)
  - dense outer layer of pyrolytic carbon (OPyC)
- Designed not to crack from thermal or mechanical stresses at temperatures **<u>up to 1600°C</u>**, and therefore can contain the *radioactive* fission products during all *conceivable* “accident” situations.



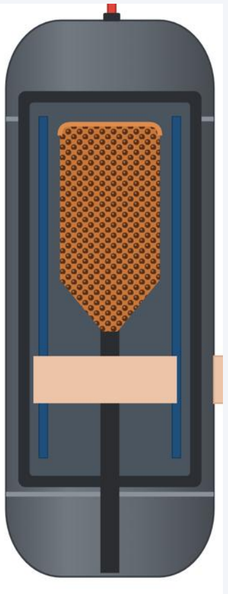
How does pebble-bed HTGR (e.g., HTR-PM) work?

Q&A



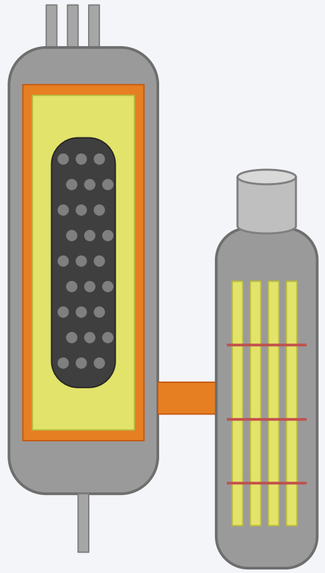
- During normal operation, helium carries heat produced in the core to the steam generator through the hot gas duct.
- Pebbles move downwards due to gravity. They are discharged at the bottom of the reactor vessel, tested and either returned to the core from the top, or sent to the spent fuel canister (continuous refueling).
- Considered to be “inherently safe” due to:
  - Strong negative feedback to change of temperature – fission stop if there is a sudden large increase in temperature.
  - Graphite reflectors have excellent heat transfer properties that enable the core to cool down to a stable and safe end-state without any operator action.
  - TRISO fuel’s ability to retain fission products below 1600°C which will not be reached in any situation.
  - Experimentally tested in reactors by shutting down helium flow to demonstrate the inherent safety – “walkaway safe”.

What is the prospect of HTGRs?



HTR-PM (China, 2 x 105 MWe)

- Criticality were achieved at the two reactors on 12 Sep and 10 Nov 2021
- Loss-of-cooling tests successfully conducted at near full power in Aug 2023
- Commercial operations in Dec 2023
- HTR-PM600 [6 modules] with 2 HPR1000 reactors commenced construction in 2026



Xe-100 (USA, 80 MWe each)

- Awarded US$80 million (initial) in US DOE Advanced Reactor Demonstration Project in Oct 2020 [up to US$1.2b federal cost-shared funding]
- Plan to deploy, with utility partner Energy Northwest, to deliver the commercial scale Xe-100 (960 MWe) by 2030
- Submitted CPA (with Dow) for 320 MWe plant at Seadrift, Texas.

In our region:

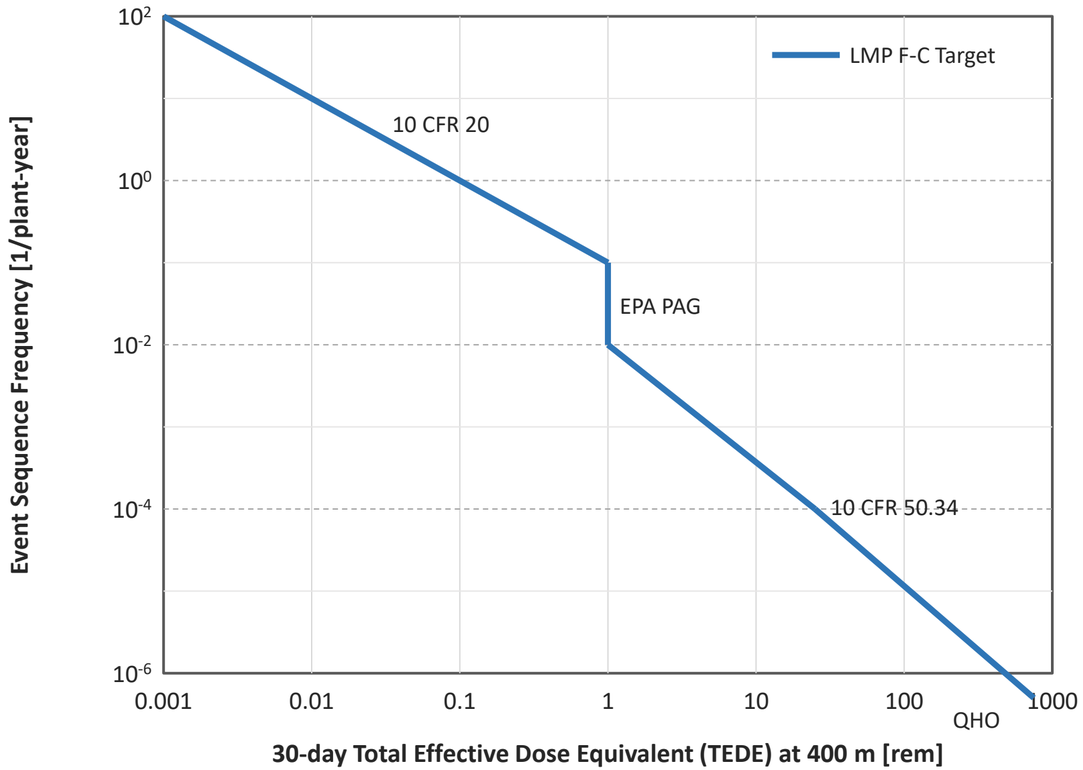
- □ Indonesia earlier initiated the process to construct an Experimental Power Reactor (RDE, 10 MWt / 3 MWe) in Serpong and also worked on a 40 MWt / 15 MWe PeLUIt-40 Reactor with Tsinghua University through their joint lab on HTGR technology.
- □ US firm Valar Atomics plans to build a micro high temperature gas reactor (100-kW, prismatic design) *with the* Philippine Nuclear Research Institute as a “proof of concept.”

How would the enhanced safety of small modular reactors affect siting?

Q&A

- Conventional large reactors requires a large "**emergency planning zone (EPZ)**" of around 15 – 30 km to ensure that population around the plant can be protected or evacuated in the unlikely event of a severe accident with large release of radioactive materials.

- For small modular reactors (including HTGR-type), due to the (1) smaller core radionuclide inventories and (2) efficient passive or "inherent" safety systems, EPZ are expected to be substantially smaller.
- Small modular reactor vendors target to reduce to **hundreds of metres or within plant site boundary** – anyone outside EPZ will not be subjected to dose above regulatory dose limits.
- *This could* allow small modular reactors to be located closer to population centres, or to co-locate with *industrial facilities or possibly* replace current coal *plants*.

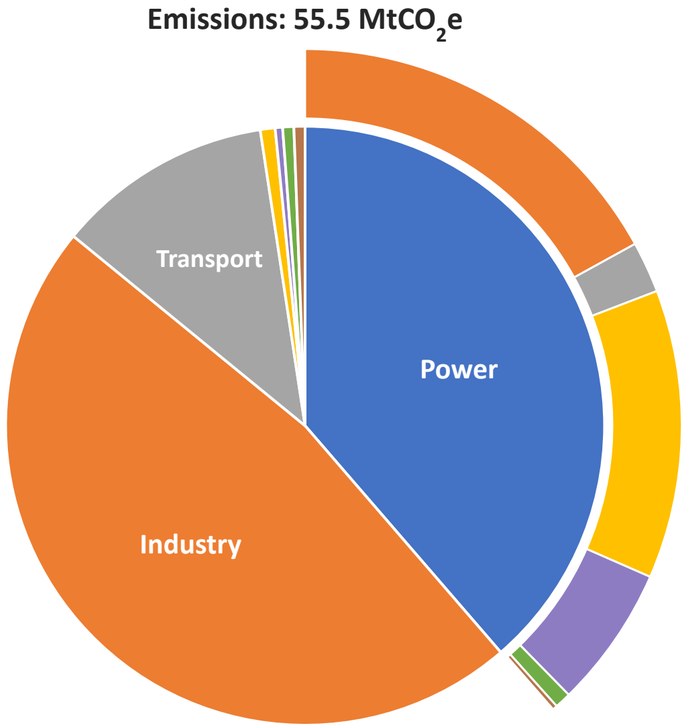


Example of EPZ determination [slightly modified from the Preliminary Safety Analysis Report (PSAR) submitted to US NRC for the construction of 4 x 80 MWe XE-100 at Long Mott Generating Station, Texas].

How can nuclear contribute beyond electricity?

- From the CO_2 emission profile of Singapore in 2023, industry contributed to 64.3% of the 55.5 $MtCO_2e$.
- Using heat directly from nuclear power reactors for the industry could potentially accelerate the decarbonization process.

- Steam from the secondary circuit after passing through the turbine in Haiyang NPP, China, have been used to heat the homes of 400,000 residents around Haiyang.
- Nuclear power has also been used in desalination projects, e.g., 14,000 m^3 of potable water per day in



PRIMARY EMISSIONS

- Power 38.7%
- Industry 47.3%
- Transport 11.7%
- Buildings 0.8%
- Household 0.4%
- Waste and Water 0.6%
- Others 0.6%

SECONDARY EMISSIONS

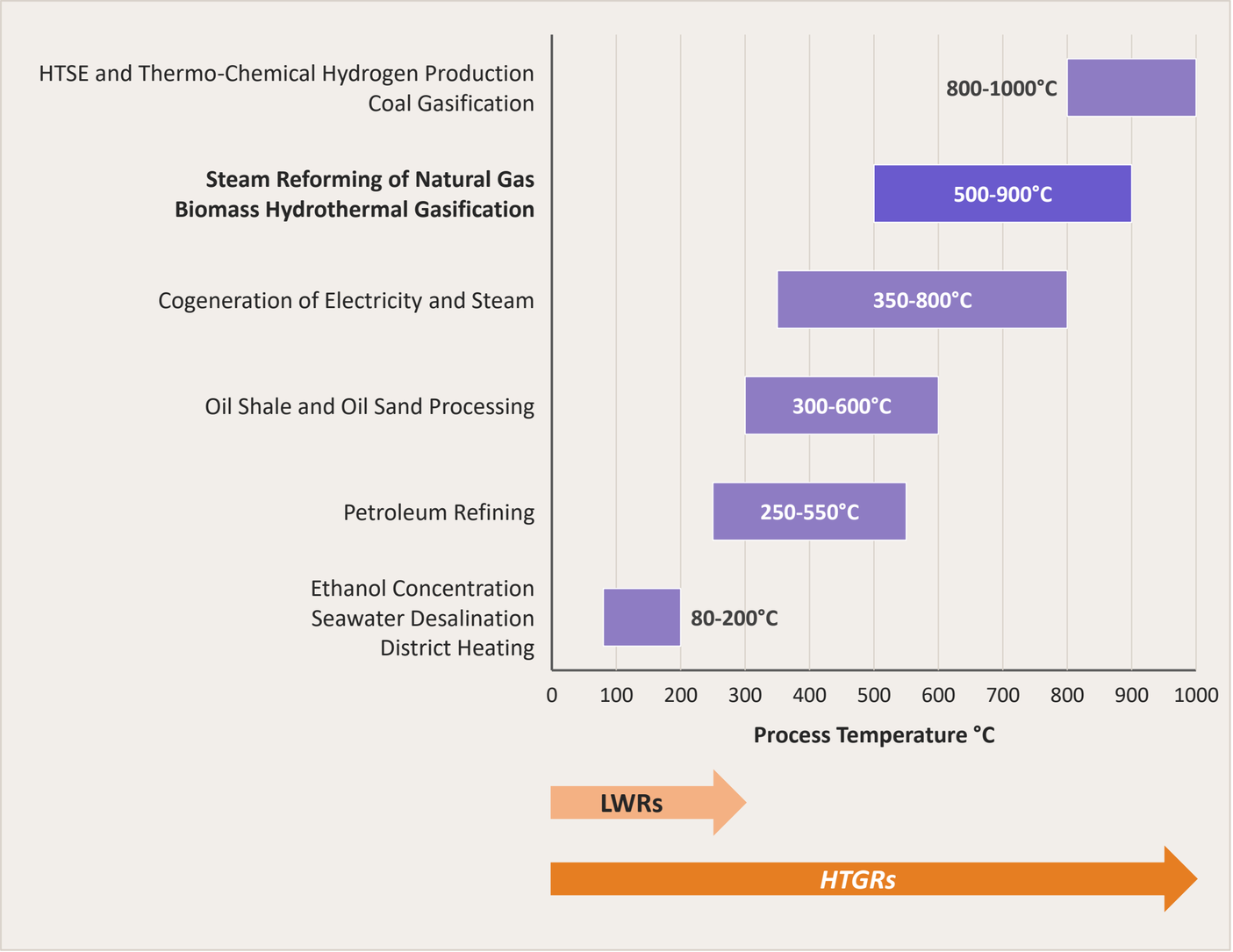
- Industry 17.0%
- Transport 2.2%
- Buildings 12.4%
- Household 6.1%
- Waste and Water 0.7%
- Others 0.2%

Primary emissions refer to direct emissions from the sector's owned or controlled sources. Secondary emissions refer to the sector's share of Power sector's emissions based on its share of electricity consumption. Outer ring: Power sector emissions allocated by electricity use.

Emission profile of Singapore in 2023

How can process heat be used in the industry?

- Higher temperature heat is needed for many other industrial processes.
- HTGRs are more suited for these process due to the higher temperature that it can produce.
- Molten Salt Reactors (with output temperature of 500 - 800°C) could also be used for industrial heat when they are commercialized.



Can HTGRs be used for industry?



Rendering of the Xuwei nuclear site, Jiangsu

- China State Council approved construction for HTR-PM600 with 2 HPR-1000 (1000 MWe PWRs) at Xuwei, Jiangsu in China in 2024.
- Construction of first HPR1000 in Jan 2026 and for HTR600 expected in Sep 2026.
- Demineralised water is heated by the primary steam of the HPR-1000s to produce saturated steam.
- The primary steam of the HTGR will then be used to heat the saturated steam for the second time to be used in the petrochemical plant (about 10 km away).
- Xuwei site is expected to contain 2 x HTR-PM600 coupled to 4 x HPR1000 and constructed in 2 phases.

https://www.youtube.com/watch?v=GSk3CKOZrC0

Will small modular reactors be too costly (per kWh) to be used?

Q&A

There are only a couple of real FOAK small modular reactors and no NOAK small modular reactors at this time to give a good real world comparison of the cost (say LCOE) of small modular reactors vs large conventional reactors. Following are points for consideration:

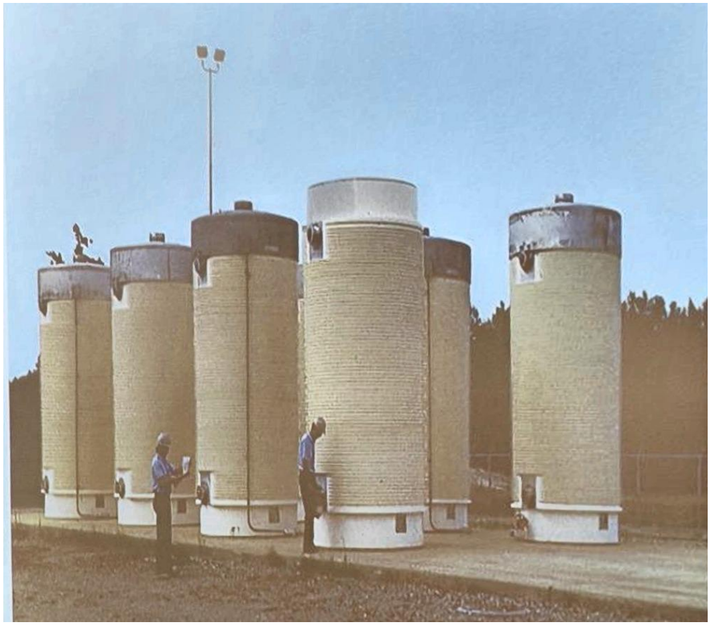
- All first-of-a-kind (FOAK) reactors are very costly due to the development cost, new supply chain, regulatory cost, etc. Cost is expected to fall significantly for the Nth-of-a-kind (NOAK) reactor as seen in various past examples in most countries.
- Further cost savings could come from manufacturing (in a factory) vs construction (at the site) due to the modular nature of small modular reactors.
- Additional cost to use conventional large reactors, e.g., grid improvement, will have to be taken into account in comparison, or for isolated communities that are not connected to grid.
- Small modular reactors built for special purpose, e.g., high temperature heat for petrochemical industry or thermochemical hydrogen production, may also be more cost effective.
- *Cost of* the nuclear island is usually a small part (typically < 20% in US) of the total cost of a nuclear power construction which includes the balance of plant (turbine, electrical yards, EPC costs, etc.).
- *With expected* shorter construction time and flexibility in deployment for small modular reactors, financing *could* be significantly easier with lower cost.

How are spent fuels managed?

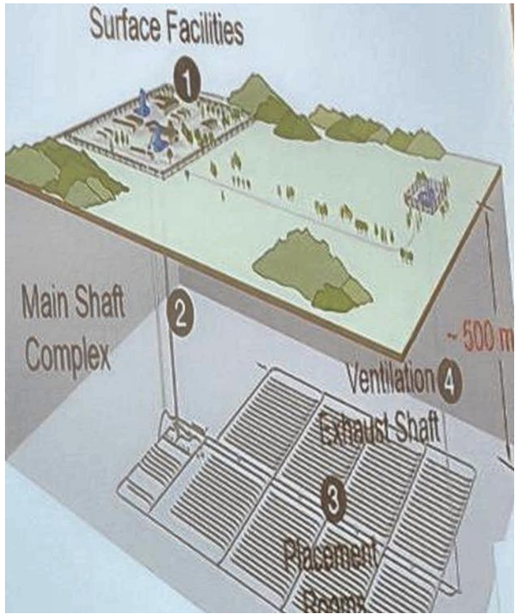
- Spent nuclear fuels are extremely radioactive, with atoms having lifetimes ranging from very short (much less than a second), to intermediate (decades) to very long (billions of years).
- Must be cooled initially, and then stored safely for a very long time.



Spent fuels are first cooled in **spent fuel pools** within the power plant.



After sufficiently cooled in SFP, the spent fuels are transferred to **dry casks** usually still within the plant.



1 Surface facilities
2 Main shaft complex
3 Placement rooms
4 Ventilation exhaust shaft
Depth ~500 m

Deep Geological Repository provides long-term storage (may be retrievable) for spent fuel. Alternative: **deep bore-holes** (up to 5 km) with *the lower* portion sealed off.

Are the wastes from small modular reactors harder to manage?

Q&A

- Nuclear waste light-water small modular reactors, such as NuScale small modular reactor or BWRX-300 will not be too different from conventional PWRs or BWRs. Volume for some wastes, such as low-level wastes could be higher due to higher level of radiation because of the smaller core.
- Wastes from non light-water small modular reactors will require different methods of treatment. HTGR spent fuel is an interesting example.
  - It has a larger volume if not treated, requiring more storage space **(-)**
  - Would be harder to be reprocessed, i.e., remove the long-lived radionuclides like uranium and plutonium **(?)**
  - It does not need to be put into a spent fuel pool after taken out of the core and can be stored as it is during intermediate stage (air cooled is sufficient) or for final disposal **(++)**

What are some key takeaways?

Q&A

- **Gen III / III+ Reactors** based on large pressurised water reactors (PWR) will continue to be the main reactors to be connected to grid in the next decade that will help in reducing CO2.
- FOAK **Small Modular Reactors** using PWR technologies with passive safety features are starting to be deployed and will very likely be commercialised in early 2030.
- **High Temperature Gas-cooled Reactor (HTGR)** technologies has been deployed and their inherently safety has been demonstrated (to a certain extent). They have great potential for supplying quality high temperature heat to relevant industries and a new plant is currently under construction in China *and* one planned in US for this purpose.
- **Hybrid systems combining nuclear and renewables** are considered by many to be *most likely* to achieve net zero emission by 2050.

What about Nuclear Fusion?

Q&A

Generation I



Big Rock Point, GE BWR

Early prototypes

- **Calder Hall** (GCR)
- **Douglas Point** (PHWR/CANDU)
- **Dresden-1** (BWR)
- **Fermi-1** (SFR)
- **Kola 1-2** (PWR/VVER)
- **Peach Bottom 1** (HTGR)
- **Shippingport** (PWR)

Generation II



Diablo Canyon, Westinghouse PWR

Large-scale power stations

- **Bruce** (PHWR/CANDU)
- **Calvert Cliffs** (PWR)
- **Flamanville 1-2** (PWR)
- **Fukushima II 1-4** (BWR)
- **Grand Gulf** (BWR)
- **Kalinin** (PWR/VVER)
- **Kursk 1-4** (LWGR/RBMK)
- **Palo Verde** (PWR)

Generation III / III+





Kashiwazaki, GE ABWR | Olkiluoto 3, AREVA PWR

Evolutionary designs

- **ABWR** (GE-Hitachi; Toshiba BWR)
- **ACR 1000** (AECL CANDU PHWR)
- **AP1000** (Westinghouse-Toshiba PWR)
- **APR-1400** (KHNP PWR)
- **APWR** (Mitsubishi PWR)
- **Atmea-1** (Areva NP-Mitsubishi PWR)
- **CANDU 6** (AECL PHWR)
- **EPR** (AREVA NP PWR)
- **ESBWR** (GE-Hitachi BWR)
- **Small Modular Reactors**
  - B&W mPower PWR
  - CNEA CAREM PWR
  - India DAE AHWR
  - KAERI SMART PWR
  - NuScale PWR
  - OKBM KLT-40S PWR
- **VVER-1200** (Gidropress PWR)

Generation IV

Safe
Secure
Sustainable
Competitive
Versatile

Arriving ~2030

Innovative designs

- **GFR** gas-cooled fast reactor
- **LFR** lead-cooled fast reactor
- **MSR** molten salt reactor
- **SFR** sodium-cooled fast reactor
- **SCWR** supercritical water-cooled reactor
- **VHTR** very high temperature reactor

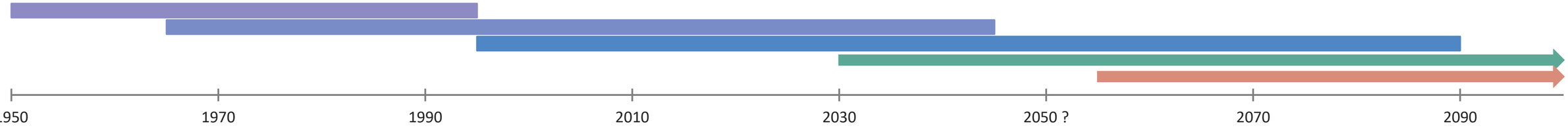
Fusion Reactors



Arriving ~2060?

Ultimate Energy Source

- **ITER / DEMO** (Tokamak)
- **EAST / CFETR** (Tokamak)
- **NIF** (Inertial Confinement Fusion)
- **CFS** (Tokamak, HTS)
- **TAE** (FRC, aneutronic)
- **GF** (Magnetized Target Fusion, piston)
- **Hyperjet** (MTF, plasma jets)



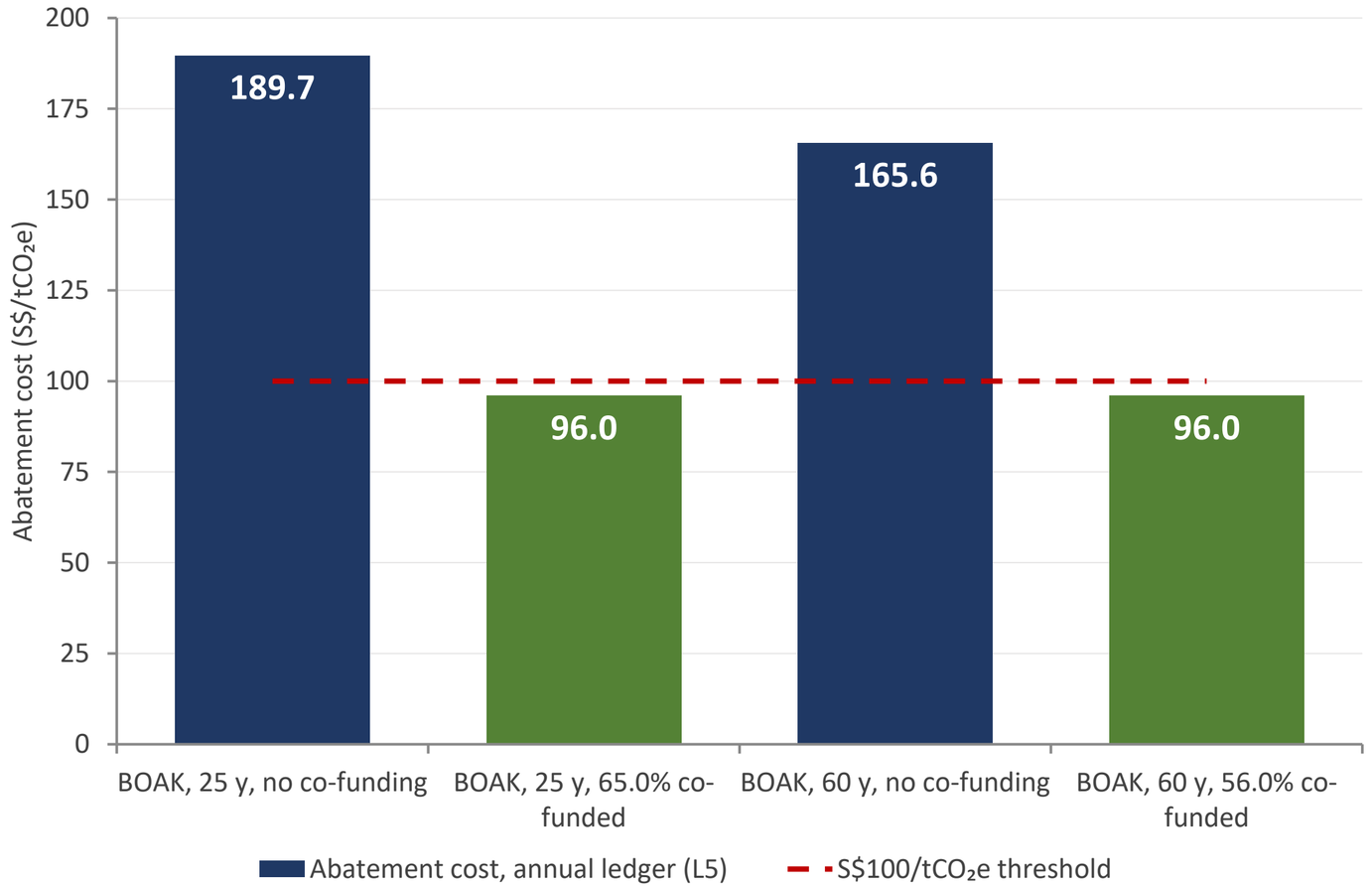
If an accident happens, where to run?



- Selected technology should have:
  - Very low frequency (say 1 in 1,000,000 reactor-years of operation) of any significant accident that could affect the public.
  - Even in the event of such an accident, the release will be low enough so that there is no evacuation or
  - There are mitigation measures to ensure no evacuation.
- That is Low frequency and Very Low Consequences!
- Answer: “No need to run”

Can government co-funding lower the abatement cost further?

Q&A



- Co-funding **S$483.0M, 65.0%** of the S$742.8M reactor capital, brings BOAK over 25 years to **S$96/tCO₂e** on the annual ledger. The cash flow needs S$483.6M, 65.1%.
- Over 60 years, **S$416.0M, 56.0%**, is enough on the ledger. The cash flow needs S$417.1M, 56.2%.
- Without co-funding, BOAK costs S$189.7 and S$165.6/tCO₂e on the ledger, and S$190.1 and S$166.0/tCO₂e by cash flow (n = 1.0, 210 MWth charged).
- Co-funding moves cost to the public budget. The total resource cost stays the same.

Government support can reduce early deployment risk, while technological maturity and industrial offtake determine long-term commercial viability.